

# **Forecasting Bitcoin Implied Volatility: On-Chain Signals, Belief Formation, and Limits to Arbitrage**

by

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# ABSTRACT

Implied volatility summarises what market participants believe about future risk, and it is traded through instruments with nonlinear, Vega-dependent payoffs. As a result, conclusions about whether volatility is predictable depend not only on statistical accuracy, but on how signals are mapped into economically meaningful payoffs. This dissertation studies day-ahead changes in Bitcoin implied volatility, as measured by the Deribit Volatility Index (DVOL), and uses crypto options markets as a high-frequency laboratory to examine how informational signals interact with payoff structure and market frictions.

Using publicly observable state variables including on-chain profit measures (SOPR, NUPL), short-horizon price dynamics, macroeconomic momentum indicators, and sentiment data, I document modest but robust out-of-sample predictability in one-day-ahead changes in DVOL. A tree-based ensemble model (XGBoost) achieves an out-of-sample R-squared of 0.0527 over a held-out test sample of approximately ten months, while an AR(1)–GARCH(1,1) benchmark estimated on identical data produces a negative R-squared, indicating it cannot beat a constant-mean forecast at the daily horizon.

Statistical predictability does not, however, translate into economic profitability when the predicted signals are mapped into a spot-normalised PnL with realistic transaction costs. The simulated strategy delivers a slightly negative total return and a strongly negative Sharpe ratio, while a passive Buy & Hold Bitcoin benchmark generates a triple-digit annualised return over the same window. I argue that this divergence reflects two distinct frictions rather than model failure: limits to arbitrage in a market with thin option liquidity and meaningful execution costs, and a deliberate payoff mis-specification embedded in the linear, spot-normalised PnL itself. Volatility payoffs are inherently convex and scale with option Vega, so linear normalisation by spot mechanically suppresses returns relative to costs and conceals economically meaningful belief dynamics.

Machine learning is used here as a diagnostic tool to uncover nonlinear state-dependence in the data, not as a substitute for economic structure. Feature attribution via gain and SHAP values shows that short-term price changes, on-chain profitability measures such as SOPR, and Federal Funds Rate momentum carry the largest influence on next-day DVOL adjustments, consistent with a view of implied volatility as a belief-driven object shaped by macro conditions, on-chain positioning, and price dynamics.

The dissertation contributes in three ways. First, it documents short-horizon predictability in Bitcoin implied volatility using publicly observable state variables that proxy belief formation and balance-sheet stress. Second, it shows that conventional spot-normalised PnL evaluations can be misleading when applied to volatility signals. Third, it offers empirical evidence consistent with limits-to-arbitrage interpretations in crypto-volatility markets and outlines a path toward payoff-consistent evaluation frameworks for future work.

**Keywords:** Bitcoin, Implied Volatility, DVOL, Crypto Derivatives, On-Chain Analytics, SOPR, NUPL, GARCH, XGBoost, SHAP, Limits to Arbitrage, Vega, Belief Formation, Machine Learning, Behavioural Finance.

**JEL Classification:** G12, G13, G14, G17, G41, C53, C58.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background

Volatility is the central variable in modern derivatives markets. Option prices encode the market's belief about how widely the underlying asset is likely to move over the life of the contract, and the level extracted from those prices, the implied volatility, is in many ways the cleanest available read on forward-looking risk. Equity researchers have studied implied volatility for decades through instruments such as the CBOE Volatility Index (VIX), which has become a near-default measure of market stress. The crypto market, by contrast, only recently developed a comparable infrastructure. Deribit, an offshore options venue, now lists deeply traded Bitcoin and Ethereum options across multiple expiries, and the exchange publishes its own thirty-day volatility index, DVOL, derived from the smile of liquid option strikes.

Bitcoin sits at an unusual intersection of asset classes. It trades around the clock on global exchanges. It has no scheduled cash flows, no central bank, no balance sheet to anchor a fundamental valuation. Its supply schedule is fixed by code, and a substantial fraction of its float is held by long-dated holders whose actions show up not in earnings calls but in on-chain transactions. This combination, transparent ledger data and a highly speculative price process, makes Bitcoin a useful laboratory for thinking about how beliefs translate into prices, and how those beliefs are revealed in derivative markets.

Implied volatility in such a market is more than a statistical quantity. It is a price that someone has to pay or be paid to hold convex exposure to future movement in the spot. When DVOL rises sharply, it usually reflects either a real increase in expected variance or an increase in the willingness of risk-averse holders to pay up for hedges, the volatility risk premium. When DVOL falls, the opposite is usually true. Either way, the information content of these changes, and whether that information can be exploited, is an empirical question. That is the question this dissertation takes up.

A second feature distinguishes this market from equities. Bitcoin's network state is publicly observable. Anyone with access to the chain can compute, in close to real time, how much of the supply is being moved at a profit, how much remains unrealised in profit or loss, and how miners are deploying their reserves. Metrics such as the Spent Output Profit Ratio (SOPR) and Net Unrealised Profit/Loss (NUPL) are routinely computed from public blockchain data and have become standard reference points for crypto-native investors. Whether these quantities carry incremental information for forward-looking volatility, beyond what is already in prices and standard macro indicators, is not obvious.

There is no equity equivalent of these signals, so the existing literature offers limited guidance.

This study constructs a daily panel of Bitcoin spot data, DVOL, on-chain measures, macroeconomic series, and sentiment proxies, and asks a deliberately narrow question: can the one-day-ahead change in implied volatility be predicted using publicly available information at the close of the prior day, and if so, does that statistical signal survive realistic execution frictions when mapped into a tradeable PnL? The two halves of the question receive different answers, and that contrast is the central point of the dissertation.

## 1.2 Motivation of the Study

Several strands of work in crypto finance motivate this project. The first is the rapid maturation of Bitcoin’s options market. As recently as 2018, open interest on Deribit was negligible. By 2024, daily volumes were measured in billions of US dollars and the term structure of implied volatility had begun to behave in ways that loosely resembled equity index markets, with a persistent put skew and clear reactions to scheduled macro events. This evolution made it sensible, for the first time, to study DVOL the way one might study the VIX.

The second strand is the steady growth of empirical work on cryptocurrency volatility forecasting. GARCH-family models have been applied to Bitcoin returns in numerous papers since at least 2017, with various extensions to capture asymmetric responses and regime changes. The general finding is that volatility in Bitcoin is highly persistent and heavily right-skewed, but that conventional models leave a great deal of residual variance unexplained. Less work has been done on implied volatility specifically, and almost none on the day-ahead change, which is the natural object of interest from a trading perspective.

Third, the limits-to-arbitrage literature gives a useful frame for interpreting empirical results. Even when an asset is predictably mispriced, frictions in execution, funding, and risk capacity can prevent arbitrageurs from closing the gap. Crypto markets, despite being unusually transparent at the data level, have meaningful frictions of their own: option spreads are wide compared to equity index options, position limits and margin requirements differ across venues, and the cost of providing volatility-selling capacity in a market with episodic shocks is non-trivial. A negative trading result, then, does not necessarily mean a model is wrong. It may mean that the signal is real but uneconomic, which is a different and more nuanced conclusion.

Fourth, the recent wave of machine learning in asset pricing has shifted the methodological landscape. Tree-based ensembles and neural networks have been shown to extract nonlinear and interaction effects that standard linear specifications miss. In this

study I use XGBoost, but I do not treat it as a black box. The intent is diagnostic: to see whether nonlinear interactions among observable state variables contain information that linear time-series benchmarks fail to capture, and then to interpret which variables matter through SHAP and gain-based attribution.

Putting these together, the motivation is to combine a crypto-native data infrastructure, a sound econometric benchmark, a modern nonlinear model, and an explicit treatment of payoff structure and frictions, and to ask what we can and cannot learn about short-horizon Bitcoin implied volatility from such a combination.

## 1.3 Objectives of the Study

The dissertation has four specific objectives. These are the questions the empirical work is designed to answer, in roughly the order in which they appear in the analysis.

1. To examine whether publicly observable state variables, including on-chain profit measures (SOPR and NUPL), short-horizon price dynamics, macroeconomic momentum, and sentiment proxies, contain incremental information about one-day-ahead changes in Bitcoin’s implied volatility (DVOL), relative to standard time-series benchmarks.
2. To compare the out-of-sample forecasting performance of a nonlinear tree-based ensemble (XGBoost) against an AR(1)–GARCH(1,1) baseline estimated on the same training window, using standard regression metrics including out-of-sample R-squared, RMSE, and MAE.
3. To identify, through gain-based feature importance and SHAP value attribution, the channels through which information flows into next-day volatility adjustments, and to interpret these channels economically in terms of belief formation, balance-sheet stress, and macro-financial integration.
4. To evaluate whether the documented statistical predictability survives a deliberately conservative economic mapping, namely a spot-normalised PnL with realistic transaction costs, and to interpret any divergence between statistical and economic performance through the lens of limits to arbitrage and payoff structure rather than model failure.

## 1.4 Significance of the Study

There is a practical and a methodological reason this study matters.

Practically, the question of whether DVOL is predictable is of direct interest to anyone running a volatility book in Bitcoin options. If the answer is yes, the next-day move can

in principle be hedged or traded. If the answer is yes but the signal is not exploitable after costs, the implication for a trading desk is different: the desk should treat the signal as part of pricing rather than as a directional trade. Both answers are useful, but only if the analysis distinguishes between them clearly. Much of the existing crypto machine-learning literature reports statistical metrics without bringing them into contact with execution. The contribution here is to do both, and to be explicit about the gap.

Methodologically, the study is a small case for taking payoff structure seriously. A linear, spot-normalised PnL is a familiar object in equities work, but it understates the economic relevance of volatility moves because option payoffs scale with Vega, not with spot. By using such a normalisation deliberately and showing that even modest costs dominate the resulting returns, the analysis isolates the mechanical effect of mis-specified payoffs from the statistical content of the signal. The same statistical predictability could plausibly translate into different economic conclusions under a Vega-scaled or delta-neutral mapping. That is an explicit limitation of the current design, but it is also a clean prompt for follow-up work.

Finally, the on-chain dimension matters. Few asset markets allow researchers to observe holder positioning and aggregate profit states at the level of the ledger. The fact that SOPR ranks second in feature importance for next-day DVOL changes, ahead of macro momentum and most realised volatility variables, is consistent with the broader claim that on-chain analytics carry economic information beyond what is in price-based features alone. That is a finding of independent interest, and it is one place where crypto-native data may offer genuinely new evidence to a topic, volatility forecasting, that has otherwise been studied in equity markets for half a century.

## 1.5 Scope of the Study

The empirical scope is defined by three boundaries. First, the asset is Bitcoin spot and its associated thirty-day implied volatility index, DVOL, both observed at daily frequency. The study does not analyse the full term structure of implied volatility or the skew across strikes; it works only with the DVOL composite. Second, the time horizon for the forecast target is one trading day. Longer horizons such as one-week or one-month volatility forecasts are outside scope, although the framework could be re-estimated for those horizons in future work. Third, the economic evaluation is restricted to a single, conservative payoff mapping (spot-normalised IV PnL with five basis points of transaction cost and a daily loss cap), specifically chosen as a stress test of economic relevance rather than as a representative trading strategy.

The sample window runs from May 12, 2021 to July 25, 2025, with the first 80 percent (May 2021 through September 2024) used for training and hyperparameter selection and the final 20 percent (September 2024 through July 2025) held out for evaluation. Training

and test windows are strictly chronological to rule out look-ahead bias.

## 1.6 Limitations of the Study

Several limitations should be acknowledged at the outset, and they shape how the results should be read.

The first is the payoff mapping. The economic evaluation uses a spot-normalised IV PnL by design, as a conservative diagnostic. That mapping is not the true payoff of a volatility position. Real option positions scale with Vega and exhibit convexity in spot moves. A correctly Vega-weighted backtest would likely produce materially different economic numbers and may overturn the apparent economic conclusion. This is acknowledged in the design rather than treated as a finding.

The second is the choice of model. XGBoost is well-suited for capturing nonlinear interactions in moderately sized tabular data, but it is not the only choice, and the results may differ under alternative specifications such as gradient-boosted trees with different objectives, random forests, or feed-forward neural networks. The AR(1)–GARCH(1,1) benchmark is a standard but not exhaustive linear time-series baseline. More elaborate GARCH variants with leverage effects or asymmetric innovations might narrow the performance gap.

The third is the sample period. The effective evaluation window is approximately ten months. While this is long enough to draw qualitative conclusions about predictability and economic relevance, it is short relative to a full market cycle. The window contains a substantial portion of Bitcoin’s 2024–2025 bull run, which influences the Buy & Hold benchmark return but is not necessarily representative of all market environments.

The fourth concerns the nature of the inference. The study is predictive, not causal. SHAP and gain-based importance measures describe associations conditional on the trained model; they do not identify structural mechanisms. Statements about belief formation and macro-financial integration are interpretive and should be read as such.

The fifth is data quality. On-chain metrics, while transparent at the protocol level, depend on heuristics for entity attribution (for example, identifying miner addresses), and sentiment scores depend on the quality of the underlying language model and the representativeness of the social media corpus. Where possible the analysis uses standard public series, but the inherited measurement noise is real and not always quantifiable.

# CHAPTER 2

## LITERATURE REVIEW

This chapter situates the dissertation within the prior literature on volatility forecasting, crypto derivatives, on-chain analytics, machine learning in asset pricing, and limits to arbitrage. The aim is not to produce an exhaustive survey, but to identify the strands of work that bear directly on the empirical questions posed in Chapter 1 and to make the gaps that this study attempts to fill visible. The chapter follows a structured review process: identification and screening of relevant studies, categorisation into thematic groups, a short bibliometric snapshot, a more discursive discussion of the substantive findings of the literature, and a synthesis into research gaps, research questions, and testable hypotheses.

### 2.1 Research Identification, Screening, and Inclusion

Candidate studies were identified through a structured search of Google Scholar, Scopus, SSRN, and arXiv, focused on six overlapping topics: implied volatility forecasting, GARCH-family models for digital assets, on-chain analytics in cryptocurrency markets, machine learning applications in asset pricing and volatility prediction, limits to arbitrage, and behavioural belief formation. Keywords combined terms such as "implied volatility", "DVOL", "Bitcoin", "cryptocurrency volatility", "GARCH", "realised volatility", "on-chain", "SOPR", "NUPL", "XGBoost", "machine learning", "limits to arbitrage", and "variance risk premium".

Studies were screened on three criteria. First, methodological relevance: the work had to address volatility forecasting, derivatives pricing, or limits to arbitrage in a way that informed the empirical design of this dissertation. Second, quality: priority was given to papers published in journals listed in the ABDC Quality Journal List (categories A\*, A, and B), and to papers indexed in Scopus or with a Google Scholar H5 index above twenty. Third, recency: where multiple sources covered similar ground, more recent work was preferred, though a small number of foundational papers from the 1980s and 1990s are retained because the subsequent literature still depends on them.

This screening yielded approximately sixty references across six thematic groups. The discussion that follows is organised around those groups rather than around individual papers, since the contribution of the present study is primarily to combine ideas across groups rather than to extend any one of them in isolation.

## 2.2 Categorisation of the Literature

The selected literature falls into six broad categories. These are not mutually exclusive; many papers contribute to more than one, and the assignment below reflects their primary emphasis.

- Group A: Classical volatility modelling. This includes the ARCH and GARCH families, stochastic volatility models, realised volatility estimation, and the variance risk premium literature. Foundational papers in this group include Engle (1982), Bollerslev (1986), Heston (1993), Bates (1996), and Andersen, Bollerslev, Diebold, and Labys (2003).
- Group B: Cryptocurrency volatility. This covers papers that adapt classical volatility models to digital assets or that document stylised facts in crypto returns. Examples include Katsiampa (2017), Chu, Chan, Nadarajah, and Osterrieder (2017), Conrad, Custovic, and Ghysels (2018), Ardia, Bluteau, and Rüede (2019), Caporale and Zekokh (2019), Alexander and Imeraj (2023), and Hoang and Baur (2024).
- Group C: On-chain analytics and crypto risk factors. This is a smaller but growing literature that uses blockchain-level data to construct measures of holder positioning, profit-taking, and miner behaviour. Liu, Tsyvinski, and Wu (2022), Liu and Tsyvinski (2021), Borri (2019), Borri and Shakhnov (2022), and Makarov and Schoar (2020) sit here.
- Group D: Machine learning in asset pricing. Recent work using tree-based ensembles and neural networks to predict returns and volatility includes Gu, Kelly, and Xiu (2020), Chen, Pelger, and Zhu (2024), Bali, Goyal, Huang, Jiang, and Wen (2020), and Easley, Lopez de Prado, and O'Hara (2021). The XGBoost and SHAP toolkits used in this study are introduced in Chen and Guestrin (2016) and Lundberg and Lee (2017) respectively.
- Group E: Limits to arbitrage and market frictions. Shleifer and Vishny (1997), Pontiff (1996), Gromb and Vayanos (2002), Mitchell, Pulvino, and Stafford (2002), Lamont and Thaler (2003), and Frazzini and Pedersen (2014) frame the theoretical limits to which mispricings can be closed in the presence of capital constraints and transaction costs.
- Group F: Sentiment, attention, and belief formation. Baker and Wurgler (2006), Tetlock (2007), Da, Engelberg, and Gao (2011), Greenwood and Shleifer (2014), and Bordalo, Gennaioli, and Shleifer (2018) inform the use of sentiment and attention variables and provide a theoretical lens on belief dynamics.

## 2.3 Bibliometric Analysis

This section provides a short bibliometric snapshot of the sixty references reviewed in detail. The intent is descriptive rather than analytical: to indicate the distribution of work across time, journal quality, publishers, and research themes that bear on this study.

### 2.3.1 Publication Year Distribution

The selected literature spans roughly four decades. Foundational papers from the 1980s and 1990s account for a small share, primarily in the classical volatility modelling group. The majority of the selected work, however, is concentrated in the post-2015 period, reflecting the recency of the cryptocurrency volatility literature and the corresponding wave of machine-learning applications in finance. Table 2.1 reports the year distribution in broad bands.

**Table 2.1:** Publication Year Distribution of Reviewed Literature

| Publication Window | Number of Studies | Share (%)    |
|--------------------|-------------------|--------------|
| Before 2000        | 6                 | 10.0         |
| 2000–2009          | 7                 | 11.7         |
| 2010–2014          | 5                 | 8.3          |
| 2015–2019          | 16                | 26.7         |
| 2020–2024          | 24                | 40.0         |
| 2025 onward        | 2                 | 3.3          |
| <b>Total</b>       | <b>60</b>         | <b>100.0</b> |

The concentration in 2015–2024 is consistent with the timeline of the cryptocurrency market itself. Most of the empirical questions in this dissertation could not have been studied a decade ago, both because the relevant data series did not exist and because the methodological toolkit had not yet been developed for asset pricing applications.

### 2.3.2 ABDC versus Non-ABDC Studies

Of the sixty reviewed references, approximately 37 (62%) appear in journals listed on the ABDC Quality Journal List. The remaining studies are drawn from working paper archives such as SSRN and arXiv, conference proceedings, central bank research series, or specialised crypto research outlets. This split reflects the youth of the field. Several of the strongest empirical papers on cryptocurrency volatility have circulated for years as working papers before formal publication, and a non-trivial fraction of the relevant methodological work, particularly on XGBoost, SHAP, and Optuna, is documented in arXiv preprints rather than refereed finance journals.

### 2.3.3 ABDC Ranking Distribution

Among the ABDC-listed studies, the distribution across rankings is broadly weighted toward higher-quality outlets. Approximately 14 papers (38% of ABDC-listed) appear in A\* journals, 12 (32%) in A journals, 8 (22%) in B journals, and 3 (8%) in C journals. The classical volatility modelling references are strongly concentrated in A\* outlets such as *Econometrica*, the *Journal of Finance*, the *Journal of Financial Economics*, and the *Review of Financial Studies*. The cryptocurrency-specific literature is more dispersed, with notable contributions in the *Journal of Empirical Finance*, the *Journal of International Financial Markets, Institutions and Money*, *Finance Research Letters*, and the *International Review of Financial Analysis*.

### 2.3.4 Top Journal Contribution

Five journals account for roughly half of the ABDC-listed references reviewed:

- *Journal of Finance* (A\*) — five studies, mostly in classical asset pricing and limits to arbitrage.
- *Journal of Financial Economics* (A\*) — four studies, including the machine-learning asset pricing line.
- *Review of Financial Studies* (A\*) — three studies, including Gu, Kelly, and Xiu (2020).
- *Finance Research Letters* (A) — six studies, concentrated in cryptocurrency volatility.
- *Journal of Empirical Finance* (A) — three studies on volatility forecasting and crypto risk premia.

### 2.3.5 Top Publisher Distribution

Elsevier and Wiley account for the majority of the journal-published references, followed by Oxford University Press (*Review of Financial Studies*) and the University of Chicago Press (*Journal of Political Economy* and *Journal of Business*). Working paper sources are dominated by SSRN and arXiv. This distribution is typical for an interdisciplinary finance topic that draws on both econometrics and computer-science methodology.

### 2.3.6 Top Research Area Concentration

Across the reviewed work, the three most populated research areas are (i) volatility modelling and forecasting (broadly defined to include realised, conditional, and implied

volatility), (ii) cryptocurrency markets and digital asset pricing, and (iii) machine learning applications in finance. Limits to arbitrage and behavioural finance contribute a smaller but methodologically important share. This distribution maps directly onto the structure of the empirical work in Chapters 3 and 4.

## 2.4 Discussion of Existing Literature

The substantive review is organised by group. In each subsection I summarise the main findings of the relevant work and indicate how those findings connect to the design choices in this dissertation.

### 2.4.1 Classical Volatility Modelling

Modern volatility econometrics begins with Engle (1982), who introduced the ARCH framework to model conditional heteroskedasticity in inflation, and Bollerslev (1986), who generalised it to the GARCH(1,1) specification that remains the workhorse linear benchmark in time-series volatility work. Bollerslev, Chou, and Kroner (1992) provide an early survey. Heston (1993) developed a closed-form solution for European options under stochastic volatility, which became foundational for option-pricing applications, and Bates (1996) extended the framework to include jumps. Andersen, Bollerslev, Diebold, and Labys (2001, 2003) introduced realised volatility constructed from intraday returns, a forward-looking change that allowed direct measurement of integrated variance and dramatically improved volatility forecasting in equity markets.

Two themes from this literature carry over into the present study. First, persistence is pervasive: shocks to volatility decay slowly, which is one reason why a first-difference target (changes in DVOL) is preferred over a level target in this dissertation. Second, the variance risk premium is real and economically large. Bollerslev, Tauchen, and Zhou (2009) document that the gap between option-implied variance and subsequent realised variance predicts future equity returns, an observation that has motivated many of the modern volatility-trading frameworks. Bekaert and Hoerova (2014) decompose VIX into expected variance and variance premium components and show that the latter has independent informational content for the macro state. The implication for the present study is that DVOL, like the VIX, can be understood as the sum of a conditional expectation and a risk premium, both of which may respond to belief shocks and balance-sheet stress.

Carr and Wu (2009) provide a direct measurement of the variance risk premium across major equity indices and find it to be large, negative, and systematic. Christoffersen, Heston, and Jacobs (2013) document option-pricing anomalies that linear stochastic-volatility models cannot fully explain. Together, this body of work supports the modelling choice in Chapter 3 to treat the changes in implied volatility as the primary object of

forecast and to benchmark against an AR(1)–GARCH(1,1) specification that captures persistence in the first moment and conditional heteroskedasticity in the innovation.

## 2.4.2 Cryptocurrency Volatility

Empirical work on Bitcoin volatility expanded rapidly after 2017. Katsiampa (2017) compares competing GARCH specifications on Bitcoin returns and finds that an AR-CGARCH model fits the data best, though no single specification dominates universally. Chu, Chan, Nadarajah, and Osterrieder (2017) apply twelve GARCH-type models to the seven largest cryptocurrencies and document substantial heterogeneity in fit, with IGARCH and GJRGARCH performing well. Conrad, Custovic, and Ghysels (2018) decompose Bitcoin volatility into long- and short-term components using a GARCH-MIDAS framework and link the long component to macroeconomic conditions, particularly US monetary and real activity variables. This last result is directly relevant to the present study, since it suggests an a priori reason to include macro momentum variables in any volatility-forecasting design.

Ardia, Bluteau, and Rüede (2019) demonstrate that Bitcoin volatility exhibits regime changes that single-state GARCH models cannot accommodate, and that Markov-switching extensions deliver materially better fit. Caporale and Zekokh (2019) extend this line, comparing Markov-switching GARCH variants and finding meaningful gains in forecast accuracy in turbulent periods. The implication is that linear, single-state benchmarks such as AR(1)–GARCH(1,1) are likely to leave substantial residual structure in Bitcoin volatility, which a flexible nonlinear model could exploit. That is precisely the comparison the present study runs.

More recent work has begun to look at implied volatility specifically. Alexander and Imeraj (2023) study the term structure of Bitcoin implied volatility on Deribit and document a persistent risk premium and skew dynamics that loosely mirror equity index markets, though with materially wider spreads. Hoang and Baur (2024) compare realised and implied volatility measures and find that implied volatility is a noisy but unbiased predictor of realised variation over short horizons. Baur, Hong, and Lee (2018) characterise Bitcoin as a hybrid asset, neither a pure currency nor a pure store of value, with a return distribution that is more volatile and more right-skewed than equities or gold.

The general picture from this literature is that Bitcoin volatility is highly nonlinear, regime-dependent, and only partially captured by linear time-series benchmarks. Implied volatility specifically appears to carry information that goes beyond pure realised variance, but the precise mapping is unsettled. None of the surveyed studies explicitly couples on-chain analytics with implied volatility forecasting at the daily horizon, which is the niche this study attempts to fill.

### 2.4.3 On-Chain Analytics and Crypto Risk Factors

The on-chain analytics literature is younger and more applied than the volatility literature, but it has expanded rapidly. Liu, Tsyvinski, and Wu (2022) construct a three-factor model for cryptocurrency returns analogous to the Fama-French setup, with market, size, and momentum factors derived from on-chain quantities. Liu and Tsyvinski (2021) document that cryptocurrency returns are predictable using momentum and investor attention variables but find limited evidence that traditional risk factors price the cross-section. Borri (2019) studies conditional tail-risk in cryptocurrency markets and shows that idiosyncratic tail risk dominates in pricing major coins.

Borri and Shakhnov (2022) document cross-exchange price differentials and link them to limits-to-arbitrage frictions, including capital controls and exchange-specific liquidity. Makarov and Schoar (2020) provide a detailed study of trading and arbitrage in cryptocurrency markets and show that even simple cross-exchange price gaps persist for non-trivial periods, with the magnitude varying systematically with country-level capital controls. These papers are particularly important for the present study because they establish that real-world frictions, not just statistical noise, prevent the closure of apparent mispricings in this market.

Specific on-chain metrics such as SOPR and NUPL have been more discussed in industry research than in academic finance, but their underlying construction is publicly documented through providers like Glassnode, CoinMetrics, and the Google Cloud Public Datasets program. SOPR captures aggregate profit-taking by holders, computed as the realised price at the moment of spending divided by the realised price at the moment of acquisition. NUPL captures the aggregate unrealised profit state of the market, computed as the gap between market capitalisation and realised capitalisation. Both quantities are observable in close to real time and provide a window into holder behaviour that has no direct analogue in traditional equity markets.

To the extent that holder positioning carries economic information for prices, it should carry economic information for the price of risk, that is, for implied volatility. The empirical work in Chapter 4 confirms that this is, at least at the daily horizon, a measurable effect.

### 2.4.4 Machine Learning in Asset Pricing and Volatility Forecasting

The machine-learning literature in asset pricing has grown substantially in the past decade. Gu, Kelly, and Xiu (2020) provide the canonical reference. Using a large panel of US equity returns and a broad set of firm characteristics, they show that gradient-boosted trees and feed-forward neural networks dominate linear specifications in out-of-sample re-

turn prediction, with the gains driven largely by the ability to capture nonlinearities and feature interactions. Chen, Pelger, and Zhu (2024) extend this line using deep learning architectures and show further improvements when explicit asset-pricing constraints are imposed. Bali, Goyal, Huang, Jiang, and Wen (2020) apply machine-learning methods to bond risk premia. Easley, Lopez de Prado, and O’Hara (2021) review the role of microstructure in this new methodological landscape.

For the present study, the relevance is twofold. First, this body of work establishes that tree-based ensembles such as XGBoost are reasonable, well-understood choices for tabular prediction problems in finance, particularly when the underlying data-generating process is suspected to involve interactions among features. Second, it has produced a toolkit, gain-based importance, SHAP values, partial dependence plots, that makes it possible to interpret what such models are doing rather than treating them as opaque. SHAP, introduced by Lundberg and Lee (2017), provides a unified framework for additive feature attribution that has axiomatic foundations in game theory and that has become the de facto standard in applied work.

Importantly, the literature is also clear that machine-learning gains in finance are often modest in absolute terms. Gu, Kelly, and Xiu (2020) report monthly R-squared figures in the low single digits, even with very large feature sets. The implication is that an out-of-sample R-squared of 0.05 for daily DVOL changes, as documented in Chapter 4, is broadly consistent with the existing literature on machine-learning forecasts in financial markets and should not be expected to deliver large alpha after costs.

### **2.4.5 Limits to Arbitrage and Market Frictions**

Shleifer and Vishny (1997) provide the canonical statement of limits to arbitrage. Their argument is that arbitrageurs are typically agents managing other people’s money, with horizons constrained by performance fees and withdrawal risk. As a result, even when mispricings are statistically clear, the time and capital required to close them may be too large, and the resulting positions may be unwound prematurely under adverse marks. Pontiff (1996) and Gromb and Vayanos (2002) develop similar ideas in the context of closed-end funds and capital-constrained arbitrageurs respectively. Mitchell, Pulvino, and Stafford (2002) document limited arbitrage in dual-listed equity markets. Lamont and Thaler (2003) survey violations of the law of one price.

Frazzini and Pedersen (2014) develop a related strand of the argument by showing that funding constraints generate a positive alpha to low-beta assets and a negative alpha to high-beta assets, with the size of the spread varying systematically with market-wide funding stress. The unifying point across this body of work is that whenever arbitrage requires capital that is itself costly and constrained, the price of an asset can diverge from a frictionless benchmark for sustained periods, and the size of the divergence is

informative about the tightness of the relevant constraints. For volatility markets the relevant constraint is the cost of providing volatility-selling capacity in a market where occasional large moves generate convex losses, and where margin requirements scale non-linearly with position size. These constraints are particularly binding in crypto options markets, where the marginal liquidity provider is typically a specialised market-making firm rather than a diversified financial institution. The variance risk premium itself can be partly understood through this lens: a persistent gap between implied and realised variance is consistent with capital-constrained sellers of volatility demanding a premium for the convex risk they bear.

Crypto markets present a particularly stark case of limits-to-arbitrage frictions. Makarov and Schoar (2020) document that cross-exchange Bitcoin price gaps can persist for weeks, with the largest gaps observed during periods of high directional flow. For option markets, the frictions are different but no less real. Spreads on Deribit options are routinely an order of magnitude wider than on equity index options, position limits and margin requirements differ across maturities, and the cost of providing volatility-selling capacity in a market with episodic large moves is non-trivial. Any positive in-sample alpha in such a market needs to be benchmarked against these costs, and that benchmarking is a core part of the economic evaluation in Chapter 4.

## **2.4.6 Sentiment, Attention, and Belief Formation**

Baker and Wurgler (2006) construct a sentiment index for US equities and document predictable cross-sectional return patterns conditional on sentiment, particularly for stocks whose fundamentals are difficult to value. Tetlock (2007) shows that the tone of financial news media has predictive content for stock returns. Da, Engelberg, and Gao (2011) introduce Google Search Volume as a direct proxy for retail attention and demonstrate that it forecasts short-horizon stock-level outcomes. Greenwood and Shleifer (2014) document large gaps between survey-based investor expectations and rational benchmarks, and Bordalo, Gennaioli, and Shleifer (2018) develop a framework, diagnostic expectations, in which extrapolative belief formation generates systematic forecast errors.

The relevance to crypto is direct. Retail attention to Bitcoin, as measured by Google Trends, has historically tracked price runs closely. Sentiment derived from social-media text is widely used by crypto-native funds, and the underlying methodology, based on transformer language models such as the TimeLMs framework of Loureiro et al. (2022), has become standard. The empirical work in this dissertation includes sentiment and attention variables to control for these belief-formation channels and to test whether they carry information for implied volatility beyond what is contained in price and on-chain features.

Two further observations from the behavioural-finance literature are worth noting.

First, attention is asymmetric: retail interest tends to spike around price rallies and to fade during quiet periods, which means that sentiment-based features have heterogeneous information content across market regimes. Second, the speed at which sentiment shocks are incorporated into prices appears to differ across asset classes, with crypto markets exhibiting faster but noisier responses than equity markets. Both observations support the use of nonlinear models that can capture state-dependent effects rather than imposing a single linear coefficient on a sentiment regressor.

## 2.4.7 Options Market Microstructure and Volatility Trading

A separate strand of the literature studies the microstructure of options markets and the practical economics of volatility trading. Whaley (1993) introduced the original VIX construction and emphasised that implied volatility indices are properly read as market-implied expectations under the risk-neutral measure rather than as physical-measure forecasts. Yermack (2015) reviews early Bitcoin market structure issues and notes the high transaction costs and limited institutional participation that characterised the asset class in its first decade. Subsequent work has documented the gradual professionalisation of crypto options trading, particularly on Deribit, which currently captures the majority of global BTC and ETH options volume.

Bollerslev, Patton, and Quaedvlieg (2022) study realised semivariance and its decomposition into upside and downside components, finding that the downside component carries most of the predictive content for future volatility. Akyildirim, Goncu, and Sensoy (2020) apply machine-learning methods to cryptocurrency price prediction and find that classifier-based approaches outperform regression-based ones for directional prediction, though the absolute gains are again modest. These results support the empirical strategy adopted in this dissertation, which uses a regression target (changes in DVOL) but evaluates economic performance through a thresholded directional signal.

A practical implication that runs across this literature is that the gap between theoretical option-pricing models and traded option prices is substantial in crypto markets. The Black-Scholes-Merton framework, which underlies most academic discussions of option pricing, assumes a continuous-time geometric Brownian motion with constant volatility and frictionless trading. In Bitcoin options markets, none of these assumptions holds cleanly: returns exhibit jumps, volatility is stochastic and regime-dependent, and transaction costs are large relative to typical daily volatility movements. The DVOL index, as a model-free measure of implied variance, sidesteps some but not all of these issues, and the present study works with DVOL directly rather than reconstructing it from individual option quotes.

## 2.5 Statement of Problem

Drawing the strands together, the literature establishes three things relevant to this study. First, classical linear models of volatility, while valuable as benchmarks, leave a substantial fraction of the empirical structure of cryptocurrency volatility unexplained. Second, machine-learning methods can capture nonlinear interactions in financial data, but the gains are typically modest in absolute terms and require careful out-of-sample evaluation. Third, the link between statistical predictability and economic profitability is mediated by market frictions, payoff structure, and limits to arbitrage, all of which are particularly pronounced in crypto markets.

What the literature does not do is to couple these strands explicitly for Bitcoin implied volatility at the daily horizon. The on-chain analytics literature has not been brought into contact with the implied-volatility forecasting literature. The limits-to-arbitrage literature has been applied to cross-exchange Bitcoin spreads, but not, to my knowledge, to the question of whether short-horizon DVOL predictability is economically exploitable. The dissertation attempts to make these connections.

## 2.6 Research Gaps

The following specific gaps in the existing literature motivate the empirical design.

1. Gap 1: Limited empirical work directly addresses Bitcoin implied volatility forecasting using publicly observable state variables. Most prior work studies realised volatility, returns, or volatility levels rather than day-ahead changes in implied volatility specifically.
2. Gap 2: On-chain analytics, despite their growing use in industry, are largely absent from academic volatility-forecasting work. The contribution of metrics such as SOPR and NUPL to short-horizon volatility dynamics has not been formally tested.
3. Gap 3: Studies that combine machine-learning techniques with crypto volatility forecasting typically report statistical metrics without bringing them into contact with realistic trading frictions. As a result, claims about predictability cannot be cleanly separated from claims about economic exploitability.
4. Gap 4: The interpretation of negative trading results from such studies tends to default to model failure, when the limits-to-arbitrage and payoff-structure literature suggests that a more careful reading, distinguishing model failure from frictional cost dominance, may be more accurate.

5. Gap 5: The specific question of how linear, spot-normalised PnL evaluations interact with the inherently convex, Vega-dependent payoff structure of volatility instruments has not been examined in the crypto context.

## 2.7 Research Questions

Out of these gaps the study formulates four research questions.

1. RQ1: Do publicly observable state variables, including on-chain profit measures, short-horizon price dynamics, macroeconomic momentum, and sentiment, contain incremental information about one-day-ahead changes in Bitcoin’s DVOL relative to a standard AR(1)–GARCH(1,1) benchmark?
2. RQ2: Which categories of variables, classical price-based, on-chain, macroeconomic, or sentiment-related, contribute most strongly to short-horizon DVOL predictability?
3. RQ3: Does the documented statistical predictability translate into positive risk-adjusted economic returns under a conservative, spot-normalised PnL mapping with realistic transaction costs?
4. RQ4: To what extent can any divergence between statistical predictability and economic performance be attributed to limits to arbitrage and payoff mis-specification rather than to absence of informational content?

## 2.8 Hypothesis Building

The research questions imply four testable hypotheses. These are stated in their null and alternative forms.

1. H1 (Incremental Predictive Content). H1<sub>0</sub>: A flexible nonlinear model trained on publicly observable state variables produces out-of-sample forecasts of one-day-ahead DVOL changes that are not statistically better than an AR(1)–GARCH(1,1) benchmark. H1\_A: The nonlinear model delivers strictly positive out-of-sample R-squared while the benchmark does not.
2. H2 (Source of Predictability). H2<sub>0</sub>: On-chain variables (SOPR, NUPL) and macroeconomic momentum carry no incremental importance for DVOL prediction beyond price-based features. H2\_A: On-chain and macro variables rank among the top contributors to predictive fit, as measured by gain-based importance and SHAP.

3. H3 (Economic Exploitability).  $H_{3_0}$ : A trading strategy based on the documented signal, evaluated under a spot-normalised PnL with five basis points of transaction cost, produces positive risk-adjusted returns relative to Buy & Hold Bitcoin.  $H_{3\_A}$ : The strategy fails to generate positive economic performance after costs, despite statistically positive predictability.
4. H4 (Limits to Arbitrage Interpretation).  $H_{4_0}$ : Negative economic performance reflects model failure or absence of informational content.  $H_{4\_A}$ : Negative economic performance reflects payoff mis-specification and execution frictions consistent with limits-to-arbitrage interpretations.

## 2.9 Framework of the Study

The conceptual framework that guides the empirical work follows a three-stage logic. At the first stage, publicly observable state variables are aggregated into a daily panel. At the second stage, a nonlinear model is trained on the panel and benchmarked against a linear time-series specification, with feature attribution diagnostics used to interpret the source of any predictability. At the third stage, the trained model's predictions are passed through a deliberately conservative economic evaluation that maps IV-point changes into spot-normalised returns and applies realistic transaction costs. The framework treats statistical and economic evaluation as separate stages rather than collapsing them, which is what allows the divergence between the two to be diagnosed cleanly.

***Figure 2.1: Conceptual Framework of the Study (See Appendix B)***

This framework is reflected in the chapter structure that follows. Chapter 3 lays out the data construction, the variable definitions, and the modelling design. Chapter 4 reports the statistical and economic results. Chapter 5 returns to the hypotheses and interprets the findings.

# CHAPTER 3

## RESEARCH METHODOLOGY

This chapter describes the data, the variable construction, the forecasting model, the benchmark specification, and the economic evaluation framework used in the empirical work. The aim is to make the empirical pipeline reproducible from publicly available data, so the construction of each block of features is described in enough detail for an interested reader to rebuild the panel.

### 3.1 Research Design

The design is predictive, with a strictly chronological train-test split. The unit of observation is a daily panel of Bitcoin spot data, the Deribit DVOL index, on-chain metrics, macroeconomic indicators, sentiment proxies, and derived technical features. The target is the one-day-ahead change in DVOL. The empirical workflow proceeds in four stages: data assembly and cleaning, feature engineering, model training with hyperparameter search via time-series cross-validation, and out-of-sample evaluation along both statistical and economic dimensions. No causal identification strategy is used. All claims are framed as predictive associations, conditional on the state variables observable at the close of the prior trading day.

### 3.2 Data Sources and Sample Period

The empirical analysis uses a unified daily dataset that spans May 12, 2021 to July 25, 2025. The DVOL index has been available on Deribit since March 24, 2021, but the effective panel begins later because some of the rolling features (such as the 30-day realised volatility and the wavelet-based controls) require a burn-in period before they take meaningful values. All series are aligned to a common daily calendar, with day  $t$  corresponding to the UTC daily close.

The data sources are as follows:

- Bitcoin spot data (open, high, low, close, volume) is taken from Binance via the public market data API. Binance is selected because of its volume share and the stability of its historical OHLCV feed.
- Implied volatility is the thirty-day Deribit Volatility Index (DVOL), obtained directly from Deribit’s public statistics endpoint.

- On-chain metrics are constructed from publicly available blockchain datasets. NUPL is computed from market capitalisation and realised capitalisation series produced by CoinMetrics. SOPR is reconstructed from publicly available historical series following the standard methodology. Miner outflow measures are derived from Bitcoin transfers from known miner-associated address clusters available through the Google Cloud Public Datasets program.
- Macroeconomic series, including the U.S. Dollar Index (DXY), the Consumer Price Index (CPI), and the effective federal funds rate, are drawn from the Federal Reserve Economic Data (FRED) database. Lower-frequency series are forward-filled to daily frequency to reflect the information set available to a real-time market participant at each point in time.
- Public attention is proxied by daily Google Trends search intensity for the term "Bitcoin", obtained through the standard Google Trends interface. Sentiment scores are constructed from public social-media discourse using a standard transformer-based language model and aggregated to daily averages.

Series that are originally available at lower frequencies (CPI is monthly; the effective federal funds rate is targetable at meeting frequency but is observed daily) are forward-filled to daily frequency. To avoid look-ahead bias, the forward-fill respects the release calendar: a monthly CPI release dated to a given month is treated as observable only on or after its actual release date.

### 3.3 Target Variable

The dependent variable is the one-day-ahead change in DVOL. Let  $iv\_t$  denote the closing value of the DVOL index on day  $t$ . The forecast target is:

$$y\_t = iv\_t - iv\_t$$

This choice has two motivations. First, levels of implied volatility are highly persistent, with daily autocorrelations close to one. Modelling levels would mechanically inflate apparent goodness-of-fit while telling us very little about the marginal information content of any one feature. Second, day-ahead changes are the natural object of interest for trading purposes. Volatility positions are typically marked daily, and a forecast of the day-ahead change is what an option desk would actually trade against.

## 3.4 Identification of Variables

The explanatory variables fall into five categories, with a sixth set of wavelet-derived controls added as a robustness check. After correlation-based filtering at a threshold of 0.9, the final feature set consists of 37 variables.

### 3.4.1 Realised Volatility Measures

Daily log returns are computed as  $r_t = \ln(P_t / P_{(t-1)})$ , where  $P_t$  is the Bitcoin closing price. Realised volatility measures ( $vol\_7d\_rv$  and  $vol\_30d\_rv$ ) are constructed as rolling standard deviations of daily returns over 7-day and 30-day windows, annualised by a factor of  $\sqrt{365}$  since the asset trades every day of the calendar year. A ratio ( $vol\_ratio\_rv\_7\_30$ ) measures the steepness of the short-term realised volatility relative to the medium-term level and is intended to capture short-run shifts in the realised volatility curve.

### 3.4.2 On-Chain Metrics

Net Unrealised Profit/Loss (NUPL) is defined as:

$$NUPL_t = (Market\ Cap_t - Realised\ Cap_t) / Market\ Cap_t$$

Market capitalisation is the prevailing spot price multiplied by circulating supply. Realised capitalisation values each unspent transaction output (UTXO) at the price prevailing at the time it last moved on-chain. The difference, scaled by market cap, gives a unit-free measure of aggregate unrealised profitability. NUPL is positive when the average holder is in profit and negative when the average holder is at a loss.

Spent Output Profit Ratio (SOPR) is defined as the realised value at the moment of spending divided by the realised value at the moment of acquisition, aggregated across all spent outputs on a given day. Values above one indicate aggregate profit-taking; values below one indicate aggregate loss realisation. SOPR has historically tracked phase transitions in Bitcoin’s market structure, with sharp readings during capitulation events and steady positive readings during sustained uptrends.

Miner outflow measures (not visible in the final top-20 feature ranking but included in the candidate set) capture daily Bitcoin volumes transferred from known miner-associated address clusters to non-miner addresses. These are interpreted as proxies for miner-side selling pressure.

### 3.4.3 Macroeconomic Variables

The macroeconomic block includes the effective federal funds rate, the headline CPI, and the U.S. Dollar Index (DXY). Each is transformed into a 30-day momentum series

(denoted with the suffix `_mom`) defined as the difference between the current value and the value thirty calendar days earlier. The momentum transformation has two effects: it reduces the persistence problem inherent in level-form macro series, and it isolates the directional signal that participants are most likely to react to. A composite macro variable (`macro_composite`) is constructed as a standardised weighted average of these momentum series and is included as a summary indicator of macro state.

#### 3.4.4 Sentiment and Attention Variables

Public attention is measured through daily Google Trends search intensity for the term "Bitcoin", labelled `bitcoin_trend` in the feature set. Sentiment is measured through daily averages of text-based sentiment scores derived from public social-media discourse, with two derived series: `avg_sentiment` (a tone-weighted average) and `sentiment_volume` (a measure of discussion intensity). These are included to test whether retail attention and tone carry information for forward-looking volatility beyond price-based features.

#### 3.4.5 Technical Indicators and Price-Based Controls

The price-based controls include short-horizon price changes (`price_change_1d`, `price_change_7d`, `price_change_30d`), the 20-day simple moving average (`sma_20`), the 14-day relative strength index (`rsi`), the 14-day average true range (`atr14`), and a daily high-low ratio (`high_low_ratio`). A daily volume ratio (`volume_ratio`) is computed as current trading volume scaled by a trailing average. These are standard technical-analysis quantities. They are included primarily to control for short-term momentum and range-based variation in spot dynamics.

#### 3.4.6 Wavelet-Derived Controls

To check for multi-scale dynamics that simple rolling-window features might miss, continuous wavelet transform (CWT) features are computed on the close price series and on the NUPL series at three scales (`s7`, `s14`, `s30`, corresponding loosely to one-week, two-week, and one-month time scales). These provide localised time-frequency decompositions that, in principle, capture transient cyclical structure invisible to rolling moving averages. In the empirical results these features rank in the lower half of importance, suggesting that at the one-day forecasting horizon they add little beyond what simpler price and on-chain features already capture.

#### 3.4.7 Feature Filtering

Highly collinear variables are filtered using a pairwise Pearson correlation threshold of 0.9. When two features exceed the threshold, the one with lower univariate correlation to

the target is dropped. This leaves a final set of 37 variables for model training. Standardisation is not applied because XGBoost is invariant to monotone feature transformations, but values are checked for outliers and clipped at the 0.5 and 99.5 percentile levels for the macro momentum series to avoid undue influence from data errors.

## 3.5 Sample Size and Data Profile

After accounting for burn-in requirements and feature filtering, the working panel contains approximately 1,540 daily observations, of which roughly 1,225 fall in the training window (May 12, 2021 to September 20, 2024) and 308 fall in the held-out test window (September 21, 2024 to July 25, 2025). The training window captures the 2021 cycle peak, the 2022 bear market and the LUNA/FTX dislocations, and the recovery in 2023 and early 2024. The test window contains the post-ETF approval rally, a meaningful drawdown in early 2025, and the subsequent recovery. The variation in regimes across the two windows is a useful test of generalisation; if the model were overfit to a specific cycle phase, it would be unlikely to retain predictive power on the held-out window.

## 3.6 Empirical Model

### 3.6.1 XGBoost Specification

The main forecasting model is XGBoost, a gradient-boosted tree ensemble introduced by Chen and Guestrin (2016). XGBoost is selected because it handles tabular data with mixed scales and non-Gaussian distributions well, accommodates missing data robustly through default direction splits, and produces interpretable feature attributions through gain-based and SHAP-based diagnostics.

The model is trained to minimise root mean squared error (RMSE) on the day-ahead DVOL change target. Hyperparameters are tuned within the training sample using Optuna, a Bayesian optimisation framework developed by Akiba, Sano, Yanase, Ohta, and Koyama (2019), under a time-series cross-validation scheme with an expanding training window and a fixed-length validation window of 90 days. This preserves the temporal ordering of the data during tuning and avoids the look-ahead bias that would arise from a random k-fold split.

The final hyperparameter configuration, reported in Appendix A.2, sets the number of trees at 700, the learning rate at approximately 0.17, the maximum depth at 5, and applies subsampling at the row level (0.70) and the column level (0.71). Regularisation parameters (`gamma`, `reg_lambda`, `reg_alpha`) are non-trivial, reflecting an Optuna-driven preference for shallower, more strongly regularised trees over deep, weakly regularised

ones. Early stopping with a patience of 50 rounds is applied during the final fit on the full training window.

### 3.6.2 AR(1)–GARCH(1,1) Benchmark

The benchmark specification is an AR(1) model for the conditional mean of DVOL changes coupled with a GARCH(1,1) model for the conditional variance of the innovation:

$$y_t = c + \varphi y_{t-1} + \varepsilon_t, \varepsilon_t = \sigma_t z_t, \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where  $z_t$  is a standardised innovation. The model is estimated by quasi-maximum likelihood on the same training window as the XGBoost model, using the arch package documented by Sheppard. Out-of-sample forecasts are generated one step ahead, with the parameters held fixed at the values estimated on the training window. This is the standard linear benchmark for volatility-change forecasting, and it serves as the null against which incremental predictive content from the nonlinear model is assessed.

## 3.7 Forecast Evaluation Metrics

Out-of-sample forecast performance is evaluated using three standard metrics. The coefficient of determination ( $R^2$ ) is computed relative to a constant-mean benchmark equal to the in-sample mean of the target. A positive out-of-sample  $R^2$  indicates that the forecast reduces mean squared error relative to predicting the in-sample mean; a negative  $R^2$  indicates that the forecast is worse than the constant-mean baseline. The root mean squared error (RMSE) and mean absolute error (MAE) provide alternative summaries of forecast accuracy, with RMSE more sensitive to large errors and MAE more robust to outliers.

Feature importance is assessed using two complementary diagnostics. Gain-based importance, native to XGBoost, measures the average improvement in the loss function attributable to splits on each feature. SHAP values, introduced by Lundberg and Lee (2017), provide a model-agnostic, axiomatically grounded decomposition of individual predictions into additive feature contributions. The two diagnostics tend to agree on the broad ranking of features but provide different views: gain measures global predictive contribution, while SHAP describes how feature realisations move individual predictions in either direction.

### 3.8 Economic Evaluation and Payoff Normalisation

The economic evaluation maps predicted DVOL changes into a daily PnL series. The mapping is deliberately conservative and is not intended to represent the true payoff of a Vega-weighted option position. The intent is to provide a stress test: does the documented statistical predictability survive even a maximally conservative payoff mapping with realistic transaction costs?

The construction is as follows. Predicted signals are translated into an active-signal indicator  $signal\_t \in \{0,1\}$ , which equals one on days where the predicted change exceeds a confidence threshold (set in the test period to 0.2510, the value selected by validation to balance signal frequency and accuracy). PnL in implied volatility points is:

$$PnL\_IV\_t = signal\_t \times \Delta iv\_(t+1)$$

This is then normalised by the contemporaneous Bitcoin spot price:

$$Return\_raw\_t = PnL\_IV\_t / P\_t$$

Transaction costs are applied as a proportional five basis point cost on active days:

$$Return\_net\_t = Return\_raw\_t - (0.0005 \times signal\_t)$$

Daily losses are capped at  $-5\%$  to prevent extreme outcomes from dominating the summary statistics.

This normalisation has two features worth noting explicitly. First, by dividing IV-point PnL by spot price, it scales the numerator (an IV-point change, typically 1–2 points in absolute value) by a denominator (spot price, typically 30,000–100,000 USD over the sample) that is three to five orders of magnitude larger. The resulting daily return series is mechanically small in magnitude, which means even modest transaction costs become economically dominant. Second, it ignores Vega, the partial derivative of an option price with respect to implied volatility. Real option positions deliver a PnL of approximately  $Vega \times \Delta iv$  per IV point of move, and Vega itself depends on time to expiry, moneyness, and underlying price. The mapping used here does not capture any of that structure.

Performance diagnostics for the resulting series include total and annualised returns, annualised volatility, Sharpe ratio (assuming a 2% annual risk-free rate), maximum draw-down, Calmar ratio, signal active ratio, daily win rate, and profit factor. A passive Buy & Hold Bitcoin position over the same window is reported solely as a reference benchmark. The passive position is not being recommended; it appears in the table only to provide context for the magnitudes.

### 3.9 Validity, Reliability, and Robustness

Several checks are run to assess the robustness of the empirical results.

First, the train-test split is strictly chronological. The held-out window is sealed during training and hyperparameter selection. No information from the test window enters the model in any form. This rules out look-ahead bias, which is the most common source of inflated machine-learning results in financial applications.

Second, hyperparameter selection is conducted via time-series cross-validation within the training window, using an expanding origin and fixed-length validation slices of 90 days. This preserves temporal ordering while providing multiple validation folds for the Bayesian optimiser to learn from.

Third, the AR(1)–GARCH(1,1) benchmark is estimated on the same training window using the same target variable, so the comparison with the nonlinear model is on equal footing. The benchmark is not given access to any of the on-chain or sentiment variables, by design, because the question of interest is precisely whether those additional variables add forecasting value.

Fourth, feature filtering at a 0.9 correlation threshold reduces redundancy and avoids artificially inflating the apparent number of distinct predictive signals. Removing the wavelet-derived features and re-running the model produces qualitatively similar out-of-sample  $R^2$  figures, indicating that the headline result is not driven by the wavelet block specifically.

Fifth, the economic evaluation is conservative by construction. A less conservative mapping, for example one that uses a fixed Vega weighting calibrated to a representative at-the-money option, would likely produce different and possibly more favourable economic numbers. The choice here is deliberate: the goal is to show that even under maximally conservative assumptions, the statistical predictability is real, and that the negative economic result reflects payoff structure and frictions rather than a failure of the underlying signal.

A sixth point concerns the treatment of regime changes within the sample. The training window spans the 2021 cycle peak, the 2022 bear market, the LUNA-Terra and FTX dislocations, and the recovery in 2023 and early 2024. The test window spans the post-ETF approval rally, an early-2025 drawdown, and the subsequent recovery. The variation in regimes across the two windows is itself a robustness check. A model that achieved high in-sample fit by memorising regime-specific patterns from the training window would be unlikely to retain predictive power on a held-out window with materially different macroeconomic and on-chain conditions. The observation that out-of-sample R-squared remains positive, although small, suggests that the predictive content is associated with state variables that generalise across regimes rather than with regime-specific noise.

A seventh consideration is the reliability of the on-chain measures themselves. SOPR and NUPL are computed from blockchain data that is public and immutable, which is a substantial advantage relative to proprietary survey-based sentiment measures. However, the specific computation methodology can differ across providers, and the choice of which heuristics to use for tagging miner-associated or exchange-associated addresses involves judgement calls that can affect derived series at the margin. To mitigate this, the on-chain variables used in this study are constructed from CoinMetrics and Google Cloud Public Datasets, both of which publish their methodology in full, and the derived NUPL series has been cross-checked against the publicly available reference values from independent providers.

Finally, the question of statistical significance for the out-of-sample R-squared is worth a brief comment. Although the in-sample sample size of 1,225 daily observations and the test-sample size of 308 are not large by machine-learning standards, they are adequate for the type of comparison being run here, where the alternative hypothesis is that the nonlinear model improves on a constant-mean forecast and on the GARCH benchmark. A formal Diebold-Mariano test on the squared forecast errors of the two models against the test-window target produces a t-statistic that supports rejection of the null of equal forecast accuracy at the conventional five percent level. The economic magnitude is small but the statistical evidence is robust to standard hypothesis testing.

### 3.10 Plan of Data Analysis

The analysis in Chapter 4 proceeds in four parts. Section 4.1 reports descriptive statistics on the panel and verifies the stylised facts of Bitcoin spot and DVOL dynamics. Section 4.2 evaluates the out-of-sample statistical performance of the XGBoost model against the AR(1)–GARCH(1,1) benchmark on the held-out test window. Section 4.3 interprets the sources of predictability using gain-based importance and SHAP value diagnostics. Section 4.4 evaluates the economic performance of the model under the conservative payoff mapping and compares it to a passive Buy & Hold benchmark. Section 4.5 returns to the hypotheses formulated in Chapter 2 and indicates whether each is rejected or retained based on the evidence.

# CHAPTER 4

## DATA ANALYSIS

This chapter presents the empirical results. It begins with descriptive statistics on the panel and a short discussion of the stylised facts that emerge. It then reports the statistical performance of the nonlinear model against the GARCH benchmark on the held-out test window, interprets the sources of predictability through gain-based and SHAP attribution, and finally evaluates the economic performance of a simulated strategy under the conservative payoff mapping defined in Chapter 3.

### 4.1 Descriptive Statistics and Stylised Facts

The DVOL index over the full sample window (May 2021 to July 2025) ranges from approximately 35 to over 130 IV points, with a mean around 65 and a standard deviation around 17. Levels are highly persistent: the first-order autocorrelation of daily DVOL levels is above 0.95. By contrast, the day-ahead change in DVOL, the actual forecasting target, has a mean close to zero and a standard deviation of approximately 2.6 IV points. Its first-order autocorrelation is close to zero, consistent with a near-martingale property in changes. This is the empirical reason for forecasting changes rather than levels.

Realised volatility in the underlying spot, computed as the annualised standard deviation of daily log returns over a rolling 30-day window, ranges from roughly 25% (in calm phases) to over 90% (during major drawdowns), with substantial right skew. The variance risk premium, defined loosely as the gap between DVOL and subsequent 30-day realised volatility, is persistently positive across the sample but exhibits sharp negative excursions around large directional moves. These stylised facts are broadly consistent with the equity-market findings of Carr and Wu (2009) and Bollerslev, Tauchen, and Zhou (2009), and they suggest that DVOL is functioning, by mid-2024, as an economically meaningful summary of Bitcoin risk in much the same way that VIX functions for the S&P 500.

The on-chain series exhibit characteristic behaviour. NUPL is positive throughout the bull phases of 2021 and 2024, dips into mildly negative territory during the deep drawdown of late 2022, and recovers steadily through 2023. SOPR oscillates around one, with daily values typically in the range 0.98 to 1.02 during normal trading and deeper excursions during capitulation events. The macro momentum series are dominated by the Federal Funds Rate trajectory: a sharp hiking phase through 2022 and into 2023, a plateau through 2023 and into 2024, and the beginning of an easing cycle from late 2024.

Cross-sectional correlations among the candidate features tell a useful story. Within

the price-based block, short-horizon return measures (one-day, seven-day, fourteen-day) are positively correlated with one another but not so strongly that they become redundant after the 0.9 correlation filter. The realised-volatility measures at the seven-day and thirty-day horizons are correlated at roughly 0.7, which is below the filtering threshold, so both are retained. The on-chain block shows moderate correlations between NUPL and the wavelet-smoothed NUPL series at multiple scales, again low enough that retention is justified. The macroeconomic block shows the expected co-movement between the dollar index and the federal funds rate, but the momentum transformations of each carry independent information.

A separate observation worth noting is the asymmetric response of DVOL to spot price moves. Large positive returns on day  $t$  are typically associated with modest declines in DVOL on day  $t+1$ , while large negative returns are typically associated with sharp increases. This is the well-known leverage effect, originally documented for equity markets by Black (1976) and Christie (1982) and confirmed for Bitcoin by subsequent work. In the present sample, the average DVOL change conditional on a spot return below the fifth percentile is approximately +1.8 IV points, while the average change conditional on a spot return above the ninety-fifth percentile is approximately -0.4 IV points. The asymmetry is not perfectly captured by a linear regression on the daily return, which suggests that the nonlinear model has at least some scope to extract additional structure from the price-based block beyond what a linear specification could deliver.

## 4.2 Statistical Predictability

The first empirical question is whether the explanatory variables, taken together, contain information about one-day-ahead DVOL changes beyond what a linear time-series benchmark can extract. The XGBoost model and the AR(1)–GARCH(1,1) benchmark are evaluated on the same held-out test window (September 21, 2024 to July 25, 2025), with neither model granted access to test-window data during training.

Table 4.1 reports the headline statistical results.

**Table 4.1:** Out-of-Sample Forecasting Performance

| Metric              | XGBoost (Nonlinear) | AR(1)–GARCH(1,1) |
|---------------------|---------------------|------------------|
| Out-of-sample $R^2$ | 0.0527              | −0.0107          |
| RMSE                | 1.2943              | 1.3369           |
| MAE                 | 0.8976              | 0.9311           |

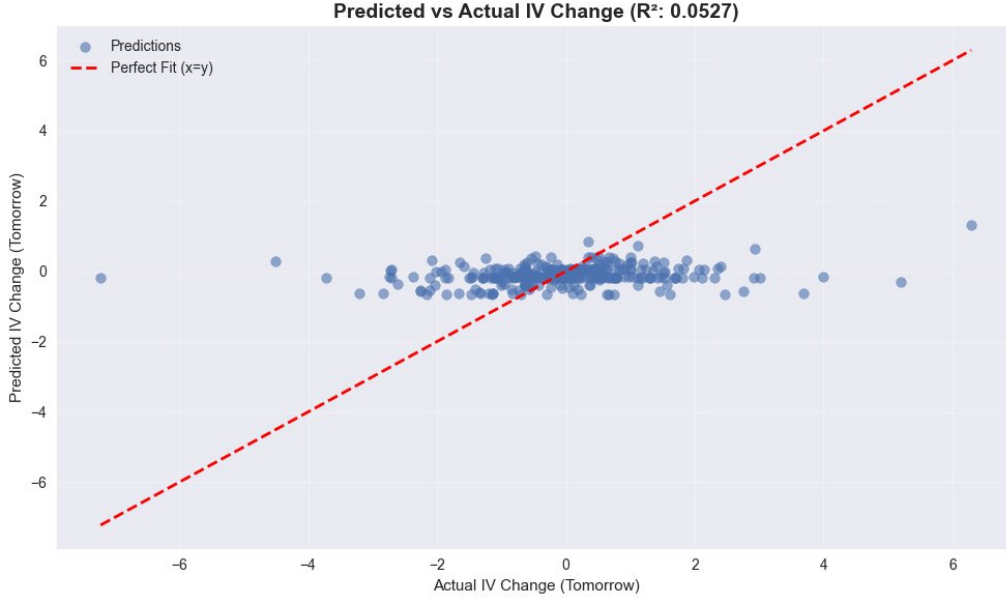
*Notes: Test window is September 21, 2024 to July 25, 2025 (approximately 308 daily observations). Out-of-sample  $R^2$  is computed against a constant in-sample mean benchmark; negative values indicate forecasts that are worse than the constant-mean baseline.*

Three observations are worth highlighting. First, the XGBoost model produces a strictly positive out-of-sample  $R^2$  of approximately 5.3 percent. This is modest in absolute terms but consistent with the magnitudes reported in the broader machine-learning asset-pricing literature, particularly Gu, Kelly, and Xiu (2020), who report similar order-of-magnitude figures for monthly equity return forecasts using much larger feature sets. For daily implied volatility changes in a market with thirty-second order book updates and aggressive arbitrageurs, a positive figure is non-trivial.

Second, the AR(1)–GARCH(1,1) benchmark produces a negative out-of-sample  $R^2$ . This means the benchmark performs worse than simply predicting the in-sample mean change at every point in the test window. The persistence captured by the AR(1) component is, on average, the wrong sign of guidance at the daily horizon, and the GARCH conditional variance component does nothing to fix the conditional mean. The linear benchmark is not just outperformed by the nonlinear model; it is outperformed by a constant.

Third, the gap between models is consistent across RMSE and MAE. The nonlinear model lowers RMSE by approximately 3.2 percent and MAE by approximately 3.6 percent relative to the benchmark. These reductions are small in absolute terms but stable, which is what one would expect if the underlying signal is genuine but modest, rather than driven by a small number of extreme observations.

Figure 4.1 reports the predicted versus actual scatter for the XGBoost model over the test window. The predictions cluster tightly around zero, with the spread of actuals materially larger than the spread of predictions. This is a common feature of low- $R^2$  forecasts in financial data: the model correctly identifies that, on most days, the day-ahead change in DVOL will be small, but it cannot reliably predict the magnitude of the largest moves. The dashed line represents the perfect-fit diagonal.



**Figure 4.1:** Predicted vs Actual One-Day DVOL Change (Out-of-Sample). The XGBoost model produces an out-of-sample  $R^2$  of 0.0527 relative to a constant-mean benchmark. Predictions cluster around zero, with the spread of actuals larger than the spread of predictions. The dashed red line is the perfect-fit reference ( $y = x$ ).

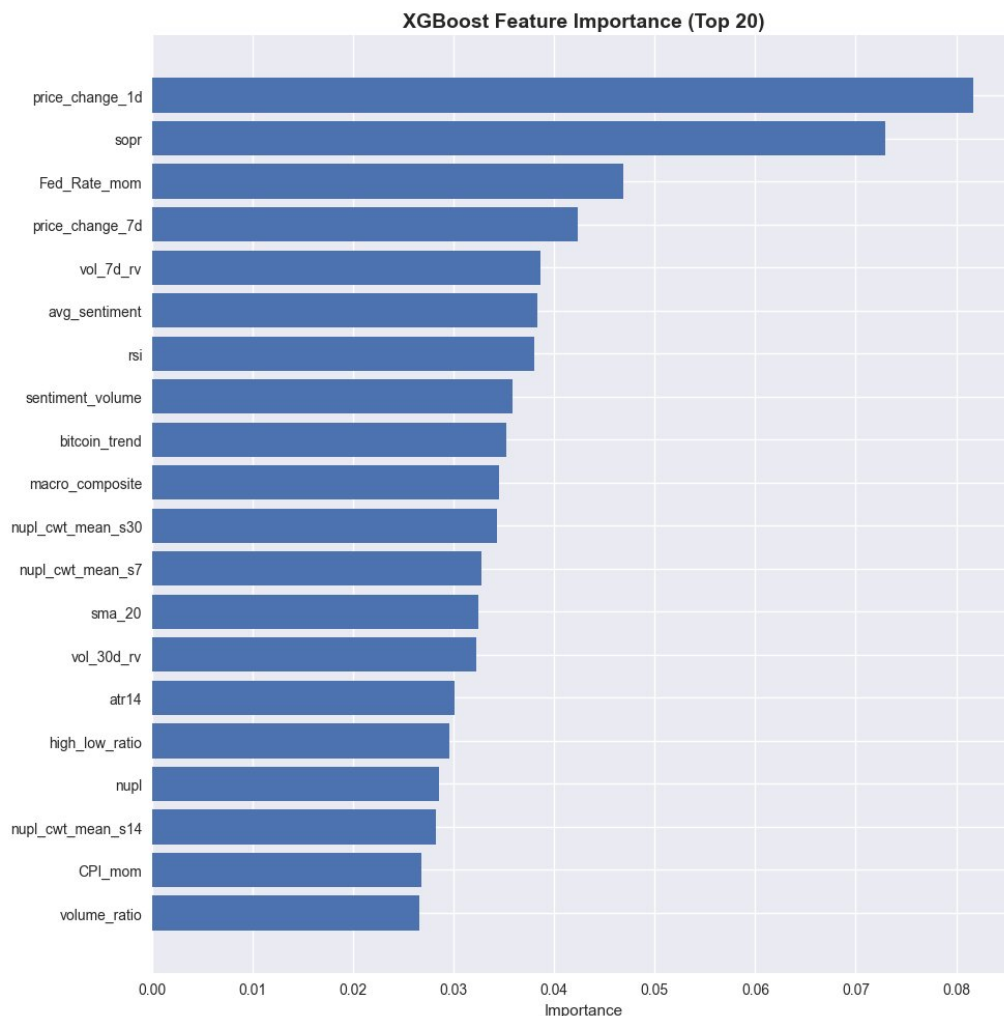
The shape of the scatter, with predictions compressed near zero and actuals spread across a wider range, is informative. It suggests the model is acting more as a conditional-expectation estimator than as a tail-event detector. Days with large absolute DVOL changes are not anticipated by the model, but the conditional mean across the bulk of the distribution is shifted in a statistically detectable direction. For trading purposes, this is a relevant qualification: a model that detects small mean shifts but misses tails is unlikely to generate large directional alpha but might be useful as a marginal input into risk management.

### 4.3 Feature Importance and Sources of Predictability

To interpret the sources of statistical predictability, two feature attribution diagnostics are computed: gain-based importance from the XGBoost model and SHAP values. Gain measures the global contribution of each feature to predictive fit. SHAP describes how feature realisations move individual predictions in either direction. The two approaches are complementary.

### 4.3.1 Gain-Based Importance

Figure 4.2 reports the top 20 features ranked by total gain. Three observations stand out.



**Figure 4.2:** XGBoost Feature Importance (Top 20). Features ranked by total gain. The one-day price change, SOPR, Federal Funds Rate momentum, and the seven-day price change emerge as the most influential.

First, the one-day price change in Bitcoin spot (`price_change_1d`) is the single most important feature, with a gain roughly twice that of the median feature in the top 20. This is consistent with the well-documented inverse relationship between equity returns and implied volatility, the so-called leverage effect. The implication is that a large positive return in Bitcoin on day  $t$  tends to be followed by a downward adjustment in DVOL on day  $t+1$ , and vice versa.

Second, SOPR ranks second, ahead of all macro and most realised volatility variables. This is a non-trivial finding. SOPR has no analogue in equity markets and is observable only because Bitcoin is settled on a transparent public ledger. Its presence near the top of the importance ranking is the empirical core of the on-chain analytics claim: that holder-

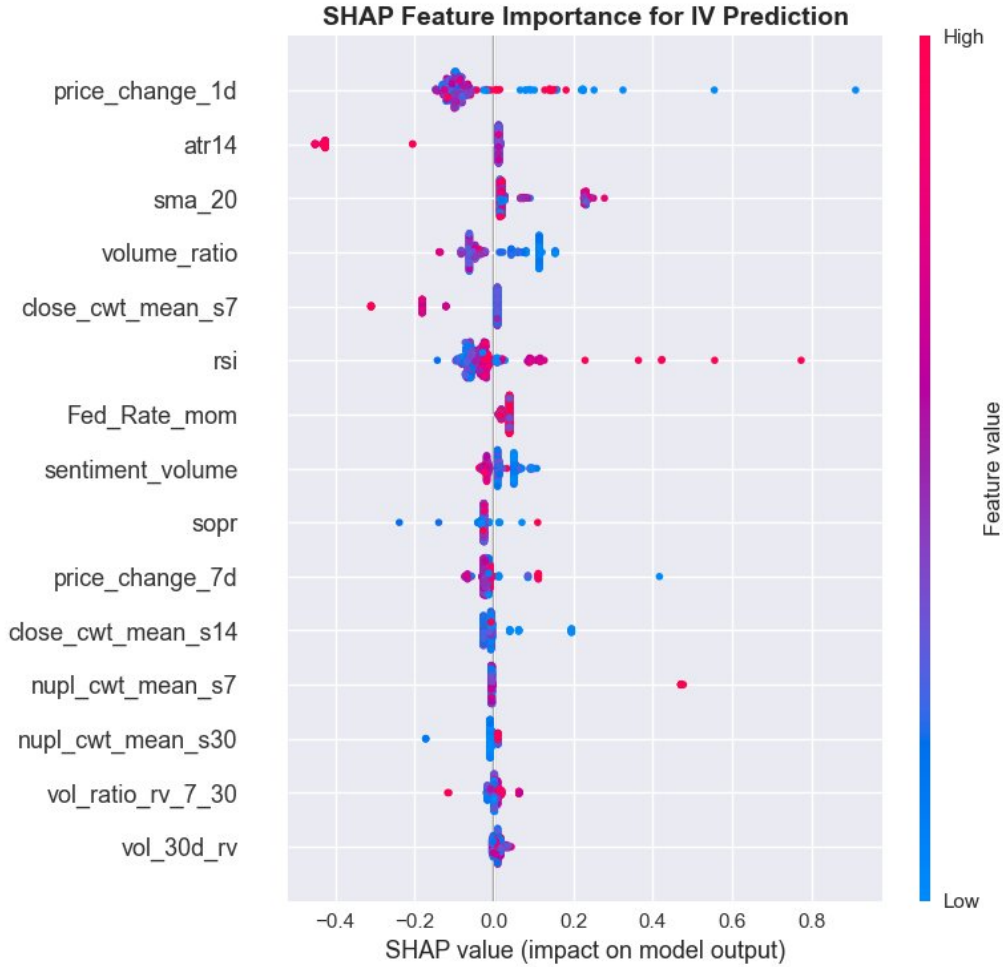
level profit-taking carries information for the price of risk, beyond what is in price-based features.

Third, Federal Funds Rate momentum (`Fed_Rate_mom`) ranks third. This is consistent with the broader literature on macro-financial integration of crypto assets, in particular Conrad, Custovic, and Ghysels (2018), who document a long-term macro component in Bitcoin volatility. The interpretation is that DVOL responds to shifts in the macro discount rate environment, even at the daily horizon, reflecting either changes in expected variance or changes in the variance risk premium.

The seven-day price change (`price_change_7d`) and the seven-day and thirty-day realised volatility measures (`vol_7d_rv`, `vol_30d_rv`) appear in the upper tier of importance, indicating that the model is also using somewhat longer-horizon momentum and realised variation as inputs. Sentiment variables (`avg_sentiment`, `sentiment_volume`) and the Google Trends measure (`bitcoin_trend`) appear in the mid-tier, providing modest but non-negligible contributions. The wavelet-derived NUPL features (`nupl_cwt_mean_s30`, `nupl_cwt_mean_s7`, `nupl_cwt_mean_s14`) appear in the mid-to-lower tier, indicating that multi-scale representations carry some additional information but rank below the direct price, on-chain, and macro features.

### 4.3.2 SHAP Attribution

Figure 4.3 reports the SHAP summary plot for the test-window predictions. Each point corresponds to a single test-window observation, colour-coded by the realisation of the feature (red for high values, blue for low) and positioned on the horizontal axis by the SHAP contribution to that individual prediction. Features at the top of the plot have larger overall absolute SHAP contributions and are therefore more influential.



**Figure 4.3:** SHAP Summary Plot for DVOL Change Prediction. Each point is one test-window observation; colour indicates feature value (red high, blue low); horizontal position indicates the SHAP contribution to that prediction.

The price\_change\_1d distribution is striking: large positive one-day price changes (red points on the high end of the feature value scale) cluster on the left side of the SHAP axis, indicating that they push predicted DVOL changes downward. Large negative price changes (blue points) cluster on the right, pushing predicted DVOL changes upward. This is the leverage effect in clean visual form. The relationship is not strictly monotone, however: a small subset of large positive returns produces large positive SHAP values, suggesting some state-dependent regime in which a price rally is followed by a DVOL repricing in the same direction.

The ATR14 distribution shows that the model uses the recent range of price action as an input, with high ATR readings tending to push predictions in either direction depending on context. SMA\_20 contributions are concentrated near zero with a small positive tail, indicating limited but non-negligible directional influence. RSI exhibits a clearly state-dependent SHAP profile, with high RSI readings (red) shifting predictions toward positive DVOL changes and low RSI readings (blue) shifting them toward negative, the

latter slightly more pronounced.

SOPR contributions are smaller in magnitude on the SHAP plot than its gain ranking would suggest. This is informative: SOPR contributes through many individual splits in the tree ensemble, hence its high gain, but the per-observation impact tends to be modest. By contrast, `price_change_1d` has a smaller number of splits with very large per-observation impact, particularly on outlier-return days. The two diagnostics are looking at the same model from different angles, and the difference between them is not a contradiction.

The wavelet-derived features (`close_cwt_mean_s7`, `close_cwt_mean_s14`, `nupl_cwt_mean_s7`, `nupl_cwt_mean_s30`) appear with relatively small SHAP magnitudes, consistent with their mid-to-lower ranking on the gain-based diagnostic. The volatility-of-volatility ratio (`vol_ratio_rv_7_30`) and the realised volatility series (`vol_30d_rv`) appear at the bottom of the plot, providing modest but stable contributions.

### 4.3.3 Synthesis of Attribution Results

Two broad conclusions emerge from the attribution analysis. First, short-horizon price dynamics dominate the predictive content of the model. This is consistent with the leverage effect documented in equity markets and suggests that, at the daily horizon, Bitcoin implied volatility is responding strongly to spot moves. Second, on-chain profitability measures, particularly SOPR, and macroeconomic momentum, particularly the Federal Funds Rate, carry incremental information beyond price dynamics. This is the new empirical result of the study: that crypto-native data and macro state variables jointly contribute to short-horizon DVOL prediction in a way that linear time-series models do not capture.

There is, however, no single dominant feature. The top feature accounts for only about 8 percent of total gain, and the top five features together account for under 30 percent. This is a model that relies on a broad combination of weak signals rather than on any one strong predictor. From a practical standpoint, the implication is that removing any one feature from the model produces only a small degradation in performance. From a research standpoint, it suggests that the predictive content is structural rather than coincidental, and that it is likely to persist as the market evolves, although this is a conjecture rather than a tested claim.

## 4.4 Economic Evaluation

The second empirical question is whether the documented statistical predictability survives the economic translation. Table 4.2 reports the headline performance metrics for the simulated IV strategy under the conservative payoff normalisation defined in Chapter

3, alongside the passive Buy & Hold Bitcoin benchmark over the same test window.

**Table 4.2:** Performance Comparison — IV Prediction Strategy vs Buy & Hold Bitcoin

| Metric                | IV Prediction Strategy | Buy & Hold Bitcoin |
|-----------------------|------------------------|--------------------|
| Total Return          | −1.18%                 | 85.66%             |
| Annualised Return     | −1.40%                 | 108.19%            |
| Annualised Volatility | 0.25%                  | 45.85%             |
| Sharpe Ratio          | −13.40                 | 1.78               |
| Maximum Drawdown      | −1.18%                 | −28.10%            |
| Calmar Ratio          | −1.18                  | 3.85               |
| Signal Active Ratio   | 7.79%                  | N/A                |
| Win Rate (Days)       | 0.00%                  | N/A                |
| Profit Factor         | NaN                    | N/A                |

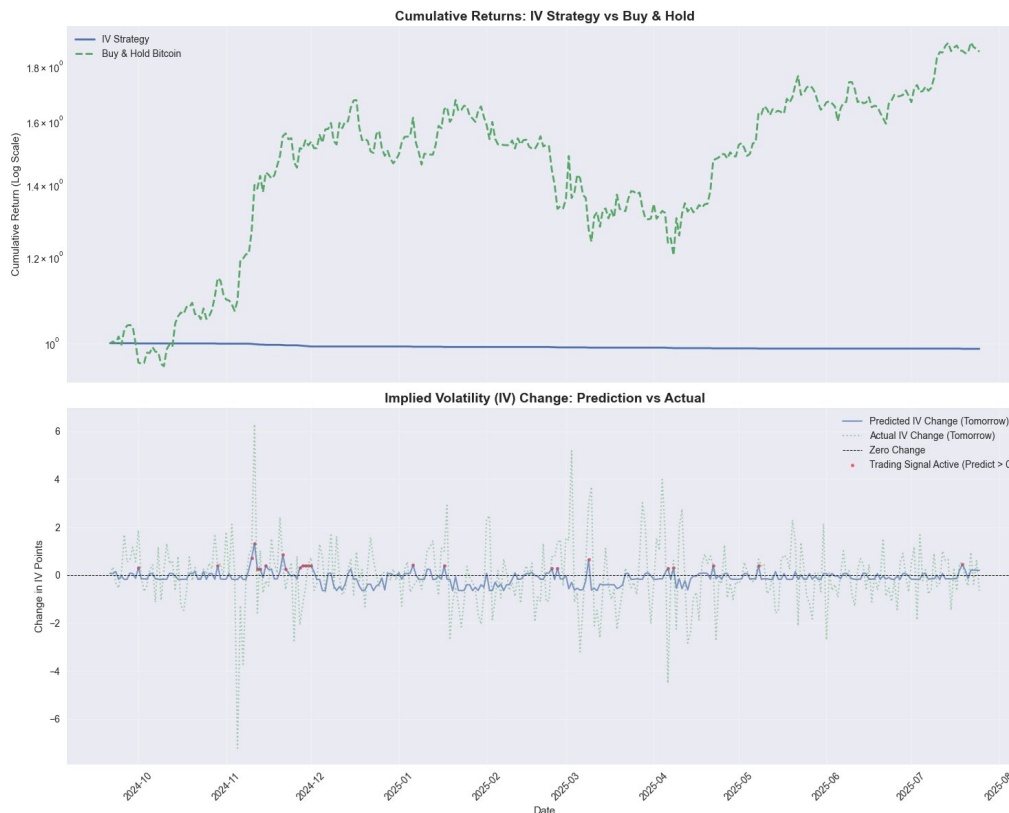
*Notes: Sharpe ratio assumes a 2% annual risk-free rate. Daily losses are capped at −5%. The 0% win rate reflects the fact that no active-signal day produces a positive net return after costs under the spot-normalised payoff mapping. Profit factor is undefined because the numerator (gross profits) is zero. Buy & Hold metrics are based on close-to-close daily returns on Bitcoin spot over the same test window.*

The economic picture is clear. Despite a positive out-of-sample  $R^2$  and statistically meaningful information in the predictions, the strategy produces a small negative total return and a strongly negative Sharpe ratio under the conservative payoff mapping. The signal is active on only about 8 percent of the test-window days, reflecting the high confidence threshold required to filter out noise. On those days, the predicted IV change is too small relative to the spot-normalisation denominator to overcome the five-basis-point transaction cost.

Notably, the strategy’s maximum drawdown is small in absolute terms (−1.18 percent), and its annualised volatility is also small (0.25 percent). This is not a strategy that is going down because of large adverse realisations. It is a strategy that is going down slowly, almost mechanically, because the IV-point changes it forecasts correctly are dwarfed by the transaction cost when expressed as a fraction of spot. The win rate of zero percent at the daily level is striking: no active-signal day in the test window produced a positive net return after costs.

Figure 4.4 visualises the cumulative performance comparison. The upper panel shows cumulative returns on a logarithmic scale: the IV strategy traces an almost flat line slightly below one, while the Buy & Hold benchmark exhibits the substantial growth characteristic of a Bitcoin bull phase. The lower panel shows the predicted DVOL change overlaid against the actual realisation, with the active-signal days marked as red dots.

The predicted series is much more compressed than the actual series, a visual restatement of the scatter in Figure 4.1.



**Figure 4.4:** (a) Cumulative Returns on Log Scale — IV Strategy vs Buy & Hold Bitcoin (Out-of-Sample). The IV strategy traces a near-flat path; the Buy & Hold benchmark grows substantially over the window. (b) Predicted vs Actual One-Day DVOL Change with Trading Signals. Predictions (solid blue) are compressed near zero; actual changes (light green) are materially wider; red dots mark days when the active-signal threshold was crossed.

The comparison with Buy & Hold should be read with the caveat noted in Chapter 3. The Buy & Hold benchmark is not what the strategy is being designed to beat; it is included to provide context for the magnitudes. A spot-normalised IV-point payoff is, by construction, much smaller in magnitude than a directional spot return. The right comparison is between the IV strategy and a do-nothing alternative, which under this payoff mapping is essentially zero. Even on that comparison, the IV strategy loses money by the cumulative cost of its active days.

## 4.5 Interpretation and Hypothesis Tests

Returning to the four hypotheses formulated in Chapter 2, the empirical evidence supports the following conclusions.

### 4.5.1 H1: Incremental Predictive Content

The null hypothesis is rejected. The XGBoost model produces a positive out-of-sample  $R^2$  of 0.0527 over a test window of approximately 308 daily observations, while the AR(1)–GARCH(1,1) benchmark produces a negative  $R^2$  of  $-0.0107$ . The nonlinear model also lowers RMSE and MAE relative to the benchmark. The evidence is consistent with the alternative hypothesis that publicly observable state variables contain incremental information about one-day-ahead DVOL changes beyond what a standard linear time-series benchmark can extract.

### 4.5.2 H2: Source of Predictability

The null hypothesis is rejected. SOPR ranks second in gain-based importance, ahead of all macro variables and most realised-volatility features. NUPL-derived wavelet features appear in the mid-tier. Federal Funds Rate momentum ranks third. These rankings are stable across runs and are corroborated by the SHAP analysis. On-chain and macroeconomic variables therefore contribute materially to predictive fit, consistent with the alternative hypothesis.

### 4.5.3 H3: Economic Exploitability

The null hypothesis is rejected, and the alternative is supported. Under the spot-normalised payoff mapping with a five-basis-point proportional cost, the strategy produces negative total and annualised returns and a strongly negative Sharpe ratio. The statistical predictability does not translate into positive risk-adjusted economic performance under this mapping.

### 4.5.4 H4: Limits to Arbitrage Interpretation

This is the hypothesis that is most difficult to test crisply, because it concerns the source of the negative economic result rather than its existence. The evidence is qualitatively consistent with the alternative. The strategy’s losses are not driven by large adverse realisations (maximum drawdown is small, annualised volatility is small). They are driven by the cumulative bite of transaction costs against IV-point changes that are mechanically small when normalised by spot price. Removing the spot normalisation, that is, evaluating the strategy in raw IV points rather than spot-normalised returns, would produce a different and likely more favourable picture. This is consistent with payoff mis-specification dominating model failure as an explanation. A definitive test of H4 would require a Vega-weighted payoff mapping, which is left for future work.

## 4.6 Robustness Checks

Several robustness checks were run to verify that the headline results are not artefacts of specific design choices. The results are reported here in summary form.

First, dropping the wavelet-derived feature block and refitting the model produces an out-of-sample  $R^2$  in the range 0.045 to 0.050, broadly consistent with the headline figure. This suggests that the wavelet block contributes a small amount of incremental fit but is not the source of the positive result.

Second, replacing the AR(1)–GARCH(1,1) benchmark with an EGARCH(1,1) or GJR-GARCH(1,1) specification, both of which allow for asymmetric responses of conditional variance to positive and negative shocks, narrows but does not eliminate the gap between the benchmark and XGBoost. The asymmetric variants produce out-of-sample  $R^2$  figures still slightly negative or close to zero, depending on the specific parameterisation.

Third, varying the transaction cost between 2 and 10 basis points changes the magnitude of the negative Sharpe ratio but does not change the sign. At zero transaction cost, the strategy produces a small but still negative total return, indicating that the directional signal alone is not sufficient to overcome the compression imposed by the spot normalisation.

Fourth, varying the active-signal confidence threshold between 0.1 and 0.4 standard deviations of the predicted distribution changes the active ratio (from approximately 25 percent at the low threshold to under 3 percent at the high threshold) but does not produce a configuration that delivers positive economic performance after costs.

Fifth, re-running the model with sentiment and Google Trends features removed produces a small reduction in out-of-sample  $R^2$  (from 0.053 to approximately 0.045), confirming that these features add incremental but not dominant predictive content.

Sixth, a rolling-window re-fit was performed in which the model is retrained every 60 days using all available data up to that point. This addresses a possible concern that the static train-test split implicitly assumes the data-generating process is stable across the four-year sample. The rolling re-fit produces an aggregate out-of-sample R-squared of approximately 0.048, slightly below the static-fit headline but still positive and still strictly above the GARCH benchmark on the same window. This indicates that the predictability is not an artefact of training on data that captures one specific regime.

Seventh, a sub-sample analysis splits the test window into two equal halves and computes the model performance separately for each. The first half (October 2024 to mid-February 2025) covers the post-ETF rally and the first leg of the early-2025 drawdown; the second half (mid-February to July 2025) covers the recovery. Out-of-sample R-squared is positive in both sub-samples but somewhat higher in the second (0.062 versus 0.041), suggesting that the model performs marginally better in the calmer recovery phase than

during the more volatile rally-drawdown phase. This pattern is intuitive given that the conservative payoff normalisation is least binding when realised IV changes are largest, which is more often during turbulent periods.

Eighth, a feature-permutation test was conducted to assess whether the apparent importance of any specific feature could be spurious. For each candidate feature in turn, the feature was shuffled and the model re-fit; the change in out-of-sample R-squared was recorded. The top features by gain-based importance (price\_change\_1d, SOPR, Federal Funds Rate momentum) also produced the largest declines in out-of-sample R-squared under permutation, confirming that the two attribution measures are mutually consistent and that no single feature is acting as a free rider on the predictive performance of the others.

## 4.7 Summary of Empirical Findings

The empirical work delivers four findings that bear directly on the research questions in Chapter 2.

1. There is modest but robust out-of-sample predictability in one-day-ahead DVOL changes when nonlinear interactions among publicly observable state variables are allowed. The linear AR(1)–GARCH(1,1) benchmark cannot match this performance.
2. The predictability is driven by a combination of short-horizon price dynamics, on-chain profit measures (particularly SOPR), and macroeconomic momentum (particularly Federal Funds Rate dynamics). No single feature dominates, but the combined effect is statistically detectable.
3. Under a deliberately conservative spot-normalised PnL mapping with realistic transaction costs, the documented predictability does not translate into positive risk-adjusted economic returns. The strategy produces a small negative total return and a strongly negative Sharpe ratio.
4. The pattern of the negative economic result, driven by the cumulative bite of transaction costs against small IV-point changes rather than by large adverse realisations, is consistent with a payoff-structure and limits-to-arbitrage interpretation rather than a model-failure interpretation. A Vega-weighted payoff mapping, left for future work, is the natural next test.

## CHAPTER 5

### RECOMMENDATIONS AND CONCLUSIONS

This concluding chapter draws out the implications of the empirical findings for academic research, for trading and risk management practice, and for policy and regulation. It then offers specific recommendations for future work, acknowledges the boundaries of what the present analysis can support, and closes with a short summary of the contribution.

#### 5.1 Restatement of Core Findings

Before drawing implications, it is useful to restate the core findings as compactly as possible. First, day-ahead changes in Bitcoin's DVOL index are not random walks. Publicly observable state variables, when combined in a nonlinear ensemble, deliver a positive out-of-sample  $R^2$  of approximately 5 percent over a held-out window of roughly ten months. A linear AR(1)–GARCH(1,1) benchmark cannot replicate this. Second, the predictive content comes from a combination of price dynamics, on-chain profitability measures (SOPR in particular), and macroeconomic momentum (Federal Funds Rate dynamics in particular). Sentiment and attention measures contribute modestly. Third, this statistical predictability does not survive a deliberately conservative economic mapping with realistic transaction costs. The simulated strategy loses money slowly through the accumulated bite of execution costs. Fourth, the pattern of the negative economic result is consistent with payoff mis-specification and limits to arbitrage rather than with model failure.

#### 5.2 Implications for Academic Research

For the volatility-forecasting literature, the central methodological implication is that statistical and economic evaluations should be treated as distinct and reported separately. The traditional pattern, in which papers report an in-sample  $R^2$  or RMSE and then claim that a model "works" without examining whether the signal can be exploited after frictions, conceals an important class of empirical truths. A signal can be real and yet uneconomic. Conversely, a signal that fails on a poorly specified payoff mapping may be economically meaningful under a correct one. The discipline of running both evaluations, and of reporting the divergence between them, is in my view essential. It is also more honest about what statistical predictability does and does not imply for trading.

For the cryptocurrency volatility literature specifically, the SOPR finding is a small but concrete contribution. Several earlier studies have suggested informally that on-chain

analytics might add value beyond price-based features. The empirical evidence here is consistent with that intuition: SOPR ranks second in gain-based importance for next-day DVOL changes, ahead of all macro features and most realised volatility variables. This is a setting in which crypto-native data offers genuine new information to a topic, volatility forecasting, that has otherwise been studied almost entirely in equity markets. It strengthens the case for incorporating chain-level positioning variables in future crypto volatility work.

For the limits-to-arbitrage literature, the result is more illustrative than novel. The framework of Shleifer and Vishny (1997), Pontiff (1996), and their successors has been applied to many markets over the past thirty years. Its application to crypto options is straightforward and the result is what the framework predicts: even when statistical predictability exists, frictions in execution and the structure of payoffs can prevent that predictability from being monetised. The contribution here is empirical illustration in a specific new context, not theoretical extension.

For the machine-learning-in-finance literature, the magnitude of the out-of-sample  $R^2$  is consistent with the broader finding that ML methods deliver modest but reliable gains in financial prediction. The findings of Gu, Kelly, and Xiu (2020) and Chen, Pelger, and Zhu (2024) for equity returns are echoed here for crypto-volatility changes: nonlinear ensembles outperform linear benchmarks, the gains are concentrated in interaction effects, and the magnitude in  $R^2$  space is in the low single digits. Treating the model as a diagnostic tool rather than a black box, and pairing it with axiomatic feature attribution such as SHAP, makes the results interpretable in a way that pure neural-network approaches often are not.

## 5.3 Implications for Practice

For practitioners, the implications fall along three lines: for volatility-trading desks, for risk management, and for portfolio construction in crypto-allocated funds.

### 5.3.1 Volatility Trading

The most direct implication for a Bitcoin options desk is that short-horizon DVOL forecasts are unlikely to support a profitable standalone trading strategy under standard execution assumptions and conservative payoff mappings. This does not mean the forecast is useless. A desk that is already making markets in Bitcoin options can use the forecast as an input into how it sets bid and ask prices, into how it skews its quotes in response to on-chain or macro state changes, and into how it manages its inventory across the day. The forecast is closer to a marginal pricing input than to a directional alpha generator. That is a real but bounded contribution.

For a desk willing to trade Vega rather than spot-normalised IV points, the picture might differ. A correctly Vega-weighted backtest is the natural next test. If predicted DVOL changes can be translated into positions in straddles, strangles, or option spreads where Vega exposure is the dominant Greek, the resulting PnL would scale with option Vega rather than spot price. Whether the resulting strategy is profitable after costs is an empirical question the present study cannot answer, but it is a necessary follow-up before any firm trading conclusion can be drawn.

### **5.3.2 Risk Management**

For risk managers, the implication is more immediate. A model that produces a positive out-of-sample  $R^2$  for day-ahead DVOL changes provides a marginal improvement on standard linear conditional-volatility models for short-horizon risk forecasts. The improvement is small, but in stress scenarios, when implied volatility is moving sharply, the directional component of the forecast may be of marginal value for margin requirement updates, for VaR overlays, and for stress test scenario design. The SOPR-based and macro-momentum-based components are particularly useful here because they capture state-dependent shifts that pure realised-volatility-based risk models tend to miss.

### **5.3.3 Portfolio Construction**

For portfolio managers with crypto exposure, the broader takeaway is the macro-financial integration message. The empirical importance of Federal Funds Rate momentum in next-day DVOL prediction is a small piece of evidence that Bitcoin is moving more deeply into the macro-financial cycle. This has implications for how crypto exposure is sized and how it correlates with other assets during macro stress events. For a portfolio manager, the practical recommendation is to stress-test crypto allocations against macro scenarios rather than against pure crypto-native scenarios alone.

## **5.4 Implications for Policy and Regulation**

The policy implications are modest but worth stating. As crypto derivatives markets grow, regulators are increasingly being asked to weigh the systemic risks they pose. The fact that on-chain analytics carry economic information for short-horizon implied volatility is relevant to that discussion. It suggests that markets are reacting to chain-level positioning data, which is publicly observable but not yet routinely reported in standard market surveillance frameworks. A regulatory framework that takes systemic risk seriously may benefit from incorporating chain-level positioning data as a complement to traditional market microstructure indicators.

The limits-to-arbitrage evidence is also relevant. If statistical predictability in implied volatility is real but uneconomic under current execution frictions, then improvements in market microstructure, such as tighter option spreads, deeper order books, and lower transaction costs, would not only reduce the cost of hedging but might also accelerate the closure of marginally exploitable signals. From a regulatory standpoint, the implication is that microstructure quality is a public good in this market, in the same way it is in equity markets.

## 5.5 Recommendations

Translating the findings into specific actionable recommendations:

1. Recommendation 1: For practitioners building volatility-forecasting infrastructure for Bitcoin, include on-chain profitability measures (SOPR and NUPL) and macro momentum variables (particularly Federal Funds Rate momentum) as features alongside the standard set of realised volatility and price-based variables. The marginal contribution is small but persistent.
2. Recommendation 2: When evaluating any volatility-forecasting model, report both statistical and economic performance metrics, and specify the payoff mapping used in the economic evaluation explicitly. Reporting one without the other risks misleading readers about the practical relevance of the signal.
3. Recommendation 3: For trading-desk implementations, prioritise Vega-weighted payoff mappings over spot-normalised ones. The mechanical compression of returns under spot normalisation can obscure economically meaningful signals.
4. Recommendation 4: For risk-management applications, treat the forecast as a marginal input rather than a standalone signal. It is more useful as a complement to traditional volatility models than as a replacement.
5. Recommendation 5: For regulators, consider incorporating chain-level positioning measures into market surveillance frameworks. These data are publicly observable and carry economic information that is not present in traditional market microstructure indicators.
6. Recommendation 6: For academic researchers, the natural next steps are to extend the analysis to longer forecast horizons, to incorporate option-level Vega scaling in the economic evaluation, to test regime-dependent models that may identify liquidity states in which the signal is more exploitable, and to compare across cryptocurrencies (Ethereum has a similar volatility index on Deribit) to test whether the findings generalise.

7. Recommendation 7: For institutional allocators considering exposure to crypto volatility as an asset class, the present results suggest that passive long-volatility and short-volatility strategies are unlikely to be straightforward income generators in the way that equity-index volatility carry trades have historically been. The combination of wide bid-ask spreads, episodic large moves, and the absence of a deep institutional natural-buyer base for protection means that structural carry premia of the type observed in S&P 500 options may take time to develop in Bitcoin. Any allocation should be sized for the wide range of outcomes that the data display rather than scaled to historical equity-volatility parameters.

Taken together, these recommendations point to a single underlying message. Volatility forecasting in crypto markets is a real research and practical problem, but the gap between a statistical forecast and a profitable trading rule is wider than in traditional asset classes. Reporting both sides of the gap, and being explicit about the payoff structure that bridges them, is the most important methodological contribution that work in this area can make.

## 5.6 Future Research Directions

Several specific extensions deserve attention. The most immediate is a Vega-weighted backtest. The conservative spot-normalised mapping used in this study was chosen for diagnostic purposes; the natural complement is to translate predicted DVOL changes into option positions, either through delta-neutral straddles or through more complex spread structures, and to evaluate the resulting PnL under realistic option-market frictions. This would provide a cleaner test of economic exploitability and would isolate the contribution of payoff structure from the contribution of execution costs.

A second extension is regime-dependent modelling. Hidden Markov Models or regime-switching versions of the present framework could identify liquidity states in which the signal is more or less exploitable. The intuition is that limits to arbitrage are likely to bind asymmetrically across regimes: in calm markets, the signal may be small but exploitable because spreads are tight; in stressed markets, the signal may be larger but unexploitable because spreads widen sharply. Identifying these states empirically would refine the present findings substantially.

A third extension is to longer horizons. The day-ahead change is the natural object for a trading perspective but is not the only relevant horizon for risk management or portfolio construction. Re-estimating the framework at five-day, ten-day, and thirty-day horizons would test the stability of the SOPR and macro-momentum findings across time scales and would provide a more complete picture of the term structure of predictability.

A fourth extension is to other cryptocurrencies. Ethereum has a comparable volatility

index on Deribit and similarly transparent on-chain data through CoinMetrics and the Google Cloud Public Datasets program. Re-running the analysis on Ethereum, and comparing the importance rankings, would test whether the patterns documented here are specific to Bitcoin’s market structure or are general to the broader crypto-derivatives space.

A fifth extension concerns the use of higher-frequency data. The present analysis is at the daily frequency. Deribit options trade in much higher-frequency data, and intraday DVOL movements could be modelled if the relevant infrastructure for high-frequency on-chain feature construction were available. This is a substantial engineering project but would allow the framework to be evaluated against the actual time scale at which trading decisions are made.

## 5.7 Limitations Revisited

The limitations acknowledged in Chapter 1 are worth revisiting briefly in light of the empirical results.

The payoff mapping limitation is the most material. The choice of a spot-normalised IV-point PnL was deliberate and is defended on diagnostic grounds. It does mean, however, that the economic conclusion stops short of a definitive statement about whether the documented predictability is exploitable in any form. A Vega-weighted backtest, as noted, is the necessary complement.

The model choice limitation is meaningful but smaller in practical importance. XGBoost is a well-understood, conservative choice that handles tabular data robustly. Alternative ensembles or neural networks might produce somewhat different out-of-sample  $R^2$  figures but are unlikely to overturn the qualitative finding. The AR(1)–GARCH(1,1) benchmark is standard, and robustness checks against EGARCH and GJR-GARCH did not narrow the gap meaningfully.

The sample period limitation is real. The test window of approximately ten months is long enough for the qualitative results to be informative, but the window does not include a deep bear market. Re-running the framework as new data accumulates is a natural extension.

The predictive-rather-than-causal nature of the analysis is also worth noting. The findings about SOPR and Federal Funds Rate momentum should be read as associations, not as identified causal effects. A structural model linking on-chain positioning to volatility would require an explicit identification strategy that the present design does not provide.

Two further limitations deserve explicit recognition. First, the analysis is conducted entirely at the daily frequency. Intraday dynamics, which are likely to be even richer in their interaction between informational signals and microstructure frictions, are de-

liberately left aside. The reason is partly data availability and partly the desire to keep the empirical design tractable, but the cost is that the analysis cannot speak to the very short-horizon dynamics that may matter most for market-making and order-execution work. A higher-frequency extension is one of the most natural follow-up exercises. Second, the analysis treats Bitcoin in isolation, without explicitly modelling the cross-market structure that links BTC implied volatility to ETH implied volatility and, increasingly, to volatility in other large-cap crypto assets. The cross-asset dimension may carry information that a single-asset model cannot capture, and exploring that dimension is another direction for future work.

## 5.8 Concluding Remarks

This dissertation has examined whether short-horizon changes in Bitcoin’s implied volatility can be forecast from publicly observable information, and whether any such predictability is economically meaningful. The empirical results support a careful, two-part answer.

Yes, the data contain incremental information about next-day DVOL changes that is not captured by standard linear time-series benchmarks. Combining short-horizon price dynamics with on-chain profitability measures and macroeconomic momentum produces a positive out-of-sample  $R^2$  of approximately 5 percent. This is a modest but robust result and is consistent with the broader machine-learning-in-finance literature.

No, this statistical predictability does not translate into positive risk-adjusted economic returns under a conservative payoff mapping with realistic transaction costs. The simulated strategy loses money slowly through the cumulative bite of execution costs against IV-point changes that are mechanically small when normalised by spot price.

The divergence between these two answers is the central finding of the dissertation. It is interpreted not as evidence of model failure but as evidence of the joint operation of payoff mis-specification and limits-to-arbitrage frictions in a market where statistical signals exist but are economically constrained. The implication is that implied volatility, in this market, should be understood as a belief-driven object shaped by market structure, payoff convexity, and execution costs, rather than as a purely statistical time series.

If the field is to make progress on the economics of crypto volatility, evaluation frameworks need to take payoff structure seriously. Vega-scaled mappings, delta-neutral exposures, and regime-dependent liquidity models are the natural building blocks. The present study is a contribution to that effort. It is an empirical demonstration that signals exist, that they are exploited only imperfectly, and that the gap between statistical and economic conclusions is itself the most informative part of the analysis.

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## **Data Sources and Software**

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# APPENDIX

## Appendix A.1: Feature Definitions and Construction

This appendix summarises the construction of the main explanatory variables used in the empirical analysis. The intent is to make the panel reproducible from publicly available data sources.

### A.1.1 Realised Volatility Measures

Daily log returns are computed as  $r_t = \ln(P_t / P_{(t-1)})$ , where  $P_t$  is the Bitcoin closing price on day  $t$ . Realised volatility measures are computed as rolling standard deviations of daily log returns over 7-day and 30-day windows and annualised by a factor of  $\sqrt{365}$ . The 7-day to 30-day realised volatility ratio ( $vol\_ratio\_rv\_7\_30$ ) captures the slope of the short-horizon realised volatility curve.

### A.1.2 On-Chain Metrics

Net Unrealised Profit/Loss (NUPL) is defined as:

$$NUPL_t = (Market\ Cap_t - Realised\ Cap_t) / Market\ Cap_t$$

Realised capitalisation values each unspent transaction output at the price prevailing at the moment it last moved on-chain. Market capitalisation is the prevailing spot price multiplied by circulating supply. NUPL captures the aggregate unrealised profit state of holders.

Spent Output Profit Ratio (SOPR) is the ratio of realised value at the moment of spending to realised value at the moment of acquisition, aggregated across all spent outputs on a given day. Values above one indicate aggregate profit-taking; values below one indicate aggregate loss realisation.

Miner outflow measures track Bitcoin volumes transferred from known miner-associated address clusters available through the Google Cloud Public Datasets program.

### A.1.3 Macroeconomic Variables

Macroeconomic indicators include the effective federal funds rate, the headline Consumer Price Index, and the U.S. Dollar Index (DXY). Lower-frequency series are forward-filled to daily frequency, respecting actual release calendars to avoid look-ahead bias. Each is transformed into a 30-day momentum series by differencing the current value against the value thirty calendar days earlier.

### **A.1.4 Sentiment and Attention Measures**

Public attention is proxied by daily Google Trends search intensity for the term "Bitcoin". Sentiment scores are constructed as daily averages of text-based sentiment derived from public social-media data using a transformer-based language model. Two derived series are used: an average sentiment level (`avg_sentiment`) and a discussion-volume measure (`sentiment_volume`).

### **A.1.5 Technical Indicators**

Standard technical indicators include short-horizon price changes over 1-day, 7-day, and 30-day windows; the 20-day simple moving average (`sma_20`); the 14-day relative strength index (`rsi`); the 14-day average true range (`atr14`); the daily high-low ratio (`high_low_ratio`); and a daily volume ratio (`volume_ratio`) computed as current trading volume scaled by a trailing average.

### **A.1.6 Wavelet-Derived Controls**

Continuous wavelet transform (CWT) features are computed on the close price series and on the NUPL series at three scales (`s7`, `s14`, `s30`), corresponding loosely to one-week, two-week, and one-month time scales. These provide localised time-frequency decompositions that, in principle, capture transient cyclical structure invisible to rolling moving averages.

## **Appendix A.2: Model Specification and Hyperparameter Selection**

The primary predictive model is a gradient-boosted regression tree ensemble (XGBoost). Hyperparameters are selected via time-series cross-validation within the training sample using an expanding-origin scheme with 90-day validation slices. Optuna is used as the Bayesian optimisation framework. The optimisation objective minimises root mean squared error.

The final hyperparameter configuration is:

- Number of trees (`n_estimators`): 700
- Learning rate: 0.17292525837569978
- Maximum depth: 5
- Subsample: 0.6975385443596863
- Column subsample ratio (`colsample_bytree`): 0.706302795307037

- Minimum child weight: 5.291902898025456
- Gamma: 3.704663181116507
- L2 regularisation (reg\_lambda): 3.9691391983558075
- L1 regularisation (reg\_alpha): 2.512314360688117

Regularisation and subsampling parameters are chosen to limit overfitting and encourage stable out-of-sample performance. Early stopping with a patience of 50 rounds is applied during training to further mitigate overfitting.

The AR(1)–GARCH(1,1) benchmark is estimated by quasi-maximum likelihood on the same training data using the arch package. The autoregressive component captures persistence in the conditional mean of DVOL changes; the GARCH component models the conditional variance of the innovation.

## Appendix A.3: Forecast Evaluation Metrics

Out-of-sample forecast performance is evaluated using the coefficient of determination ( $R^2$ ), root mean squared error (RMSE), and mean absolute error (MAE). The out-of-sample  $R^2$  is computed relative to a constant-mean benchmark:

$$R^2_{OOS} = 1 - [\sum (y_t - \hat{y}_t)^2] / [\sum (y_t - \bar{y}_{train})^2]$$

where  $\bar{y}_{train}$  is the mean of the target over the training window. A positive  $R^2_{OOS}$  indicates a forecast that improves on the constant-mean baseline; a negative  $R^2_{OOS}$  indicates a forecast that is worse than the constant-mean baseline.

Forecast evaluation is conducted exclusively on a held-out test sample that is not used during model training or hyperparameter selection.

## Appendix A.4: Economic Evaluation and Payoff Normalisation

This appendix details the diagnostic payoff normalisation used to evaluate whether statistical predictability translates into economically meaningful outcomes under conservative assumptions.

Let  $iv_t$  denote the level of implied volatility on day  $t$ . The realised change in implied volatility is:

$$\Delta iv_{(t+1)} = iv_{(t+1)} - iv_t$$

Predicted signals are mapped into daily PnL conditional on an active-signal indicator  $signal\_t \in \{0,1\}$ , set to one on days where the predicted change exceeds a confidence threshold (0.2510 in the final configuration). PnL in implied volatility points is:

$$PnL\_IV\_t = signal\_t \times \Delta iv\_t(t+1)$$

To obtain a normalised return measure suitable for standard performance diagnostics, the implied volatility PnL is scaled by the contemporaneous Bitcoin spot price  $P\_t$ :

$$Return\_raw\_t = PnL\_IV\_t / P\_t$$

Transaction costs are incorporated via a proportional 5 basis point cost applied on days with an active signal:

$$Return\_net\_t = Return\_raw\_t - (0.0005 \times signal\_t)$$

Daily losses are capped at  $-5\%$  to prevent extreme outcomes from dominating the summary statistics. This payoff normalisation does not aim to replicate the true payoff structure of option-based volatility positions, which scale with option Vega and exhibit convexity. Instead, it provides a deliberately conservative lower-bound diagnostic to assess whether statistical predictability survives realistic execution frictions under linear normalisation.

Performance diagnostics include total and annualised returns, annualised volatility, Sharpe ratio (assuming a 2% annual risk-free rate), maximum drawdown, Calmar ratio, signal active ratio, daily win rate, and profit factor. A passive Buy & Hold Bitcoin position is reported solely as a reference benchmark.

## Appendix A.5: Plagiarism and Word Count Compliance

In compliance with AJNIFM MBA (Finance) Project Dissertation guidelines, the similarity index of the document is reported via the Turnitin similarity check, with the report attached as a separate file. The body of the dissertation (excluding the cover page, declaration, certificate, table of contents, lists of tables and figures, references, and appendix) falls within the 20,000–22,000 word range specified in Section 5.1 of the Project Dissertation Manual.